A small-diagonal transfer from a degree-33 Fermat character to the quartic-CM determinant of a Prym eightfold, and the Hodge conjecture for its powers under explicit quotient hypotheses

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Abstract

We study one explicit quartic-CM case of the Hodge conjecture for abelian varieties that lies beyond imaginary-quadratic Weil-class methods. Let $\alpha = (7, 10, 13, 19, 22, 28)$ be a degree-33 Fermat fourfold character and B_8 the associated eight-dimensional ramified Prym factor of a cyclic-cover Jacobian. Unconditionally, using exact Frobenius computations at two primes, we show $\mathrm{End}^0_{\mathbf{Q}}(B_8) = E = \mathbf{Q}(\zeta_{33})^{\langle 4\rangle}$ and $Hg(B_8) = \mathrm{Res}_{E^+/\mathbf{Q}} SU_E$ with $E^+ = \mathbf{Q}(\sqrt{33})$. A structural theorem for balanced special-unitary abelian varieties then gives $\mathrm{Hdg}^2(B_8)_{\mathbf{Q}} = D^2(B_8) \oplus W_E(B_8)$, where $W_E(B_8) = \bigwedge^4_E H^1(B_8, \mathbf{Q})$, and shows that the classical and generalized Hodge conjectures for every self-power B_8^n are equivalent to algebraicity of $W_E(B_8)$. The source Fermat block is algebraic by the level-66 lift of Jumagulov (arXiv:2608.18134, Theorem 1.4), and alternatively by a cubic-to-Fermat correspondence elaborating the cubic-scroll strategy of Favero–Kelly. Our main contribution is an explicit transfer: a common finite quotient of a Kummer curve product and the Fermat character quotient, a Poincaré curve-to-Prym divisor correspondence, and contraction along the small diagonal of B_8 give a single Chow correspondence carrying the source block onto $W_E(B_8)$, with exact normalization 5^{-3} relative to canonical descent of coefficients from $\mathbf{Q}(\zeta_{33})$ to E. This avoids the semiregularity step in secant-sheaf approaches. The main theorem is stated as a single chain: four explicit quotient hypotheses H1–H4 imply algebraicity of $W_E(B_8)$, which implies the rest; our arguments for H1–H4 are given separately. Under H1–H4, $W_E(B_8)$ is algebraic and the classical and generalized Hodge conjectures hold for all B_8^n. The exact computations are supplied. We do not claim the Hodge conjecture for X^4_{33} or in general.

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Problem and contribution

Determinant subspaces attached to quartic CM fields lie beyond the reach of imaginary-quadratic Weil classes, and the Hodge conjecture for them is open in general. This paper studies one explicit such case, coming from the degree-33 Fermat fourfold character $\alpha = (7, 10, 13, 19, 22, 28)$ and an eight-dimensional ramified Prym factor B_8 of the associated cyclic-cover Jacobian, with $E = \mathbf{Q}(\zeta_{33})^{\langle 4\rangle}$ and $W_E(B_8) = \bigwedge^4_E H^1(B_8, \mathbf{Q})$.

Unconditional results. (i) $\mathrm{End}^0_{\mathbf{Q}}(B_8) = E$ and $Hg(B_8) = \mathrm{Res}_{E^+/\mathbf{Q}} SU_E$ with $E^+ = \mathbf{Q}(\sqrt{33})$ (Section 1, by exact computation). (ii) $\mathrm{Hdg}^2(B_8)_{\mathbf{Q}} = D^2(B_8) \oplus W_E(B_8)$, and for every n the Hodge ring of B_8^n is generated by divisor classes and W_E-determinant classes (Section 7); hence the classical, and via Abdulali, generalized Hodge conjecture for all B_8^n is equivalent to algebraicity of $W_E(B_8)$. (iii) The full source block $M_\alpha \subset H^4(X^4_{33})$ is algebraic, by a cubic-to-Fermat graph correspondence and independently by the level-66 lift of Jumagulov [28, Theorem 1.4], which closes the block by the Lefschetz $(1,1)$ theorem, da Silva's contraction and Aoki's Theorem 1-4 (Sections 3 and 6.2).

Conditional result. Assuming the quotient hypotheses H1–H4 stated in Section 2, an explicit global small-diagonal Chow correspondence T_B transfers the source block onto $W_E(B_8)$, with exact normalization 5^{-3} relative to canonical Galois descent (Sections 4–6). Consequently, under H1–H4, $W_E(B_8)$ is algebraic and the classical and generalized Hodge conjectures hold for every self-power B_8^n.

Dependency chain. The main theorem has the single form

$$\text{H1–H4} \implies W_E(B_8) \text{ algebraic} \implies \text{classical and generalized Hodge conjectures for all } B_8^n.$$

The second implication is unconditional (Sections 7–8). For the first, Sections 2 and 4 contain our arguments for H1–H4, and Sections 4–6 prove that H1–H4 imply the algebraicity of $W_E(B_8)$. We keep H1–H4 as explicit hypotheses in the statement, so that a reader can check them separately; we would welcome such checks.

Introduction

The algebraicity of Weil-type Hodge classes on abelian varieties has recently been settled in dimension four: Markman proved that the Hodge–Weil classes are algebraic on every abelian fourfold of Weil type [1,2], which, combined with the structure theorem of Moonen and Zarhin [3,4], establishes the Hodge conjecture for all abelian varieties of dimension at most four; Floccari, Fu and Varesco further obtained the Hodge conjecture for all powers of Weil fourfolds of discriminant one [5,6]. In dimension eight, and for determinant subspaces attached to a *quartic* CM field rather than to an imaginary quadratic one, no such result is available. The purpose of this paper is to formulate and test a finite-quotient and Abel–Fourier transfer for one explicit eight-dimensional model.

Two recent OpenAI preprints bear on this setting. The first [7] gives a lemma supplying determinant lines for faithful characters of cyclic covers, and claims the rational Hodge conjecture for self-powers of smooth labelled diagonal complete intersections, a scope that includes X_{33}^4; its Theorem 1.1 treats Jacobians of abelian covers at tensor-Hodge-generic points of the marked family, which our curve is not, since the order-five automorphism ρ gives a rational $(1,1)$-tensor that does not stay Hodge on the whole family (Section 9). The second claims the Hodge conjecture for all CM abelian varieties (preprint). Combined with that CM-Hodge claim and the ordinary operations passing from a curve to an isogeny factor of its Jacobian, the determinant supply of [7] would also have consequences for $W_E(B_8)$. The useful distinction is that the present argument does not use the CM-Hodge claim, which we have not checked; we use neither preprint as an input.

Our contribution is threefold and is deliberately confined to a single fixed object. First, for the degree-33 Fermat character $\alpha = (7, 10, 13, 19, 22, 28)$ we determine unconditionally the geometric endomorphism algebra, the Neron–Severi rank and the Hodge group of the associated ramified Prym eightfold B_8, obtaining $\mathrm{End}^0_{\mathbf{Q}}(B_8) = E = \mathbf{Q}(\zeta_{33})^{\langle 4\rangle}$ and $Hg(B_8) = \mathrm{Res}_{E^+/\mathbf{Q}} SU_E(V,h)$; this rests on two Frobenius polynomials at the split good primes 331 and 661 together with an independent number-field elimination, both computed exactly and included in the supplement. Second, under the source and finite-quotient inputs H1–H4, we construct a correspondence chain showing that the full rational quartic-CM determinant space $W_E(B_8) = \bigwedge^4_E H^1(B_8, \mathbf{Q})$ is spanned by codimension-two algebraic cycle classes. The Fermat block M_α is supplied by the level-66 lift of Jumagulov [28, Theorem 1.4] (Lefschetz, da Silva and Aoki closure; the argument is reproduced in Section 3), and alternatively by the cubic-to-Fermat correspondence of Section 6.2, which elaborates the cubic-scroll strategy of Favero–Kelly [23, §4.3] with a corrected cubic equation. The latter maps the full primitive cubic $(2,2)$-space onto M_β, then a coordinate permutation identifies M_β with M_α. The chain then uses descent through the finite quotient $X_{66}^4 \to X_{33}^4$, a common finite quotient of a Kummer curve product and the Fermat character quotient, and a Poincare curve-to-Prym correspondence whose fourfold external product is contracted along the small diagonal. The exact finite-quotient pull–push formula makes the cohomological composition scalar on the invariant block, replacing a Chow-level composition calculation. The all-powers classical and generalized Hodge conjectures are recorded only as conditional consequences of that transfer.

We claim novelty only for this fixed character and this fixed model, and only for the concrete realization just described. The passage from algebraicity on a Fermat source to algebraicity of a determinant substructure on the associated generalized Prym variety is not new: it is the mechanism of Schoen's reduction for cyclic covers of $\mathbf{P}^v$ branched along $v+2$ hyperplanes [8], and our branch data merely fall outside the quasi-simple hypothesis under which Schoen's Theorem 3.4 concludes directly. Likewise, the structural statement of Section 7, which describes the Hodge ring of every self-power in terms of divisor classes and mixed determinant classes, is a special case of the standard Weil-class framework

of Moonen–Zarhin [3,4], recorded here with a complete proof because we need its precise integral-polarization form; related statements appear in work of Milne [9] and of Varesco [6]. The mechanism of Section 8 is likewise not new: it is the Hazama–Abdulali paradigm deducing the generalized Hodge conjecture from the classical one for all self-powers [10,11,12], applied here by direct citation. What does not appear in the literature, to the best of our knowledge, is a complete proof of the algebraicity of the full rational determinant space of a *ramified* Prym eightfold with commutative quartic CM multiplication; the present manuscript proves that it follows from the explicit quotient hypotheses H1–H4.

The abelian varieties of Fermat type to which B_8 belongs were introduced and their complexified Hodge rings studied algorithmically by Shioda [13,14]; the present paper works inside that class but with a ramified Prym model and a non-full CM endomorphism field.

1. The fixed objects

Let

$$C : y^{33} = \prod_{i=0}^{4} (z - \zeta_5^i)^{a_i}, \qquad (a_0, \dots, a_4) = (7, 28, 13, 19, 10),$$

be the normalized Kummer curve, and let ρ be the order-five automorphism satisfying

$$\rho \delta \rho^{-1} = \delta^4,$$

where δ generates the order-33 deck group. Put

$$c_{33}(j) = \sum_{u \in (\mathbf{Z}/33)^\times} \zeta_{33}^{uj}, \qquad e_{33} = \frac{1}{33} \sum_{j=0}^{32} c_{33}(j)\, \Gamma_{\delta^{-j}} \in CH^1(C \times C)_{\mathbf{Q}}. \tag{1.0}$$

The coefficients are the integer Ramanujan sums $20, 1, -2, -10$, according as $j \equiv 0$, $\gcd(j, 33) = 1$, $3 \mid j$ but $11 \nmid j$, or $11 \mid j$ but $3 \nmid j$. On a δ^*-eigenline with eigenvalue ζ_{33}^n, the correspondence $\Gamma_{\delta^{-j}}$ acts by ζ_{33}^{jn}; hence (1.0) acts by

$$\frac{1}{33} \sum_{u \in (\mathbf{Z}/33)^\times} \sum_{j=0}^{32} \zeta_{33}^{j(n+u)} = \begin{cases} 1, & \gcd(n, 33) = 1, \\ 0, & \gcd(n, 33) > 1. \end{cases}$$

This Fourier calculation on all characters of $\langle \delta \rangle$ shows that the corresponding element of $\mathbf{Q}[\langle \delta \rangle]$ is idempotent; composition of graph correspondences realizes it as a Chow projector. Thus e_{33} is the rational Chow projector onto the exact-conductor- 33 part of $H^1(C)$; it kills $H^0(C)$ and $H^2(C)$, and is self-transpose. Since $c_{33}(4j) = c_{33}(j)$, conjugation by ρ fixes (1.0), so e_{33} commutes with ρ as a Chow correspondence. Put

$$D = C/\langle \rho \rangle, \qquad J = J(D), \qquad K = \mathbf{Q}(\zeta_{33}), \qquad E = K^{\langle 4 \rangle}.$$

Let $B_8 \subset J$ be the exact-conductor abelian factor defined by

$$p_B = \frac{1}{5} q_* e_{33} q^*, \qquad q : C \to D.$$

The E-compatible isogeny splitting is written

$$i : B_8 \hookrightarrow J, \qquad r : J \to B_8, \qquad ri = [N]_{B_8}, \qquad \pi_B = N^{-1} i r.$$

Set

$$U = \bigwedge_{\mathbf{Q}(\zeta_{33})}^{4} e_{33} H^1(C, \mathbf{Q}), \qquad U^\rho = U^{\langle \rho \rangle}, \qquad W_E(B_8) = \bigwedge_{E}^{4} H^1(B_8, \mathbf{Q}).$$

The exterior-power description of U is Schoen's construction [15, Lemma 1.2]. Both spaces displayed above are four-dimensional over $\mathbf{Q}$.

The notation $W_E(B_8) \subset H^4(B_8, \mathbf{Q})$ uses the canonical determinant realization. For each embedding $\sigma : E \hookrightarrow \mathbf{C}$, put $H_\sigma = H^1(B_8, \mathbf{Q}) \otimes_{E,\sigma} \mathbf{C}$. Then

$$W_E(B_8)_{\mathbf{C}} = \bigoplus_{\sigma : E \hookrightarrow \mathbf{C}} \bigwedge_{\mathbf{C}}^{4} H_\sigma \subset \bigwedge_{\mathbf{C}}^{4} H^1(B_8, \mathbf{C}) \simeq H^4(B_8, \mathbf{C}),$$

and the sum descends to the rational subspace denoted $\bigwedge_E^4 H^1(B_8, \mathbf{Q})$.

Endomorphisms and the Hodge group

Write

$$V = H^1(B_8, \mathbf{Q}), \qquad E = \mathbf{Q}(\zeta_{33})^{\langle 4\rangle} = \mathbf{Q}(\sqrt{-3}, \sqrt{-11}), \qquad E^+ = \mathbf{Q}(\sqrt{33}).$$

The graph of the deck transformation δ acts on the exact-conductor block, and $\rho\delta\rho^{-1} = \delta^4$. Hence the fixed field of $\langle 4\rangle$ commutes with the order-five descent and gives algebraic correspondences

$$E \hookrightarrow \mathrm{End}^0_{\mathbf{Q}}(B_8). \tag{1.0b}$$

We normalize the K-action by $\zeta_{33} \mapsto \Gamma_\delta$ in the covariant correspondence algebra. On $H^1(C)$, this graph acts as $(\delta^{-1})^*$; the embedding in (1.0b) is the induced action of $E = K^{\langle 4\rangle}$ on the exact-conductor, ρ-invariant factor. Transposition sends the graph of δ to that of δ^{-1}. The Rosati involution induced by the polarization therefore restricts to complex conjugation on E: the conductor and order-five averaging projectors are stable under transposition, so the graph relation passes to the factor B_8.

We first record the finite-field lemma used to show that (1.0b) is an equality.

Lemma 1.1. Let $A/\mathbf{F}_q$ be an abelian variety of dimension g, let $P_A \in \mathbf{Q}[X]$ be its Frobenius polynomial, and let π be a root. Suppose that P_A is irreducible of degree $2g$ and that the quotient of two distinct roots of P_A is never a root of unity. Then

$$\mathrm{End}^0_{\overline{\mathbf{F}}_q}(A) = \mathbf{Q}(\pi).$$

Proof. For every $r \geq 1$, the conjugates of π^r remain distinct. Thus $[\mathbf{Q}(\pi^r) : \mathbf{Q}] = 2g$ and $\mathbf{Q}(\pi^r) = \mathbf{Q}(\pi)$. Tate's theorem [16] identifies $\mathrm{End}^0_{\mathbf{F}_{q^r}}(A) \otimes \mathbf{Q}_\ell$ with the centralizer of π^r on the $2g$-dimensional Tate module. This centralizer is $\mathbf{Q}_\ell[\pi]$. Every geometric endomorphism is defined over a finite extension of $\mathbf{F}_q$, so taking the union over r proves the claim. □

We apply the lemma at the split good primes $q = 331, 661$. Both are congruent to one modulo 165, so the branch points, the order-five action, and the exact-conductor projector have good reduction. We fix the primes of the defining field by the reductions $\zeta_5 \mapsto 64 \bmod 331$ and $\zeta_5 \mapsto 247 \bmod 661$; other primitive fifth roots of unity can give different Frobenius polynomials. Norm-compatible Kummer character sums on the reduction of the displayed curve, followed by Poincare duality, give Frobenius polynomials of the form

$$P_q(X) = X^{16} + \sum_{k=1}^{7} a_{q,k}\left(X^{16-k} + q^{8-k}X^k\right) + a_{q,8}X^8 + q^8, \tag{1.1a}$$

where

q	$a_{q,1}$	$a_{q,2}$	$a_{q,3}$	$a_{q,4}$	$a_{q,5}$	$a_{q,6}$	$a_{q,7}$	$a_{q,8}$
331	1	761	−13183	147512	−9668365	99077160	−1981264536	64505725132
661	6	454	9182	716346	23307588	314636724	16817734524	266633006035

(1.1b)

The finite-field computation in (1.1a)–(1.1b) is the geometric input: it computes the Frobenius action on the exact-conductor, order-five invariant factor defining B_8. Starting from these two polynomials, a separate exact number-field calculation verifies irreducibility and factors

$$R_q(Y) = \mathrm{Res}_X(P_q(X), P_q(YX)).$$

For both primes the factor-degree multiset is

$$1^{\times 16}, \quad 16, \quad 24^{\times 4}, \quad 32^{\times 4}, \tag{1.1c}$$

and the only cyclotomic factor is $Y - 1$. Lemma 1.1 therefore gives

$$\mathrm{End}^0_{\overline{\mathbf{F}}_q}(B_{8,q}) = K_q := \mathbf{Q}(\pi_q), \qquad q = 331, 661. \tag{1.1d}$$

The complete subfield computation gives one, three, one, one, and one subfields of degrees $1, 2, 4, 8, 16$, respectively, in each K_q. The unique quartic subfield has discriminant 1089 and is isomorphic to E, for which $X^4 + 28X^2 + 64$ is a defining polynomial. The remaining discriminants needed below are

q	$\mathrm{disc}(K_{q,8})$	$\mathrm{disc}(K_q)$
331	$3^5 11^4\, 17243383\, 2429661660185017$	$3^{10}11^8\, 17243383^2\, 2429661660185017^2$
661	$2^4 3^4 11^4 107^2 227\, 199931\, 604727\, 2548061$	$2^8 3^8 11^8 107^4 227^2 199931^2 604727^2 2548061^2$

(1.1e)

Here $K_{q,8}$ denotes the unique degree-eight subfield. The Sage computation listed under Reproducibility produces (1.1a)–(1.1b) from the reduction of the curve, with $\zeta_5 \mapsto 64 \bmod 331$ and $\zeta_5 \mapsto 247 \bmod 661$, and checks that the result equals the displayed polynomials; the PARI/GP computation begins with those polynomials and proves (1.1c)–(1.1e). Thus their geometric and number-field roles are distinct.

Let $A = \mathrm{End}^0_{\mathbf{Q}}(B_8)$. Good-reduction specialization and (1.1d) give $E \subseteq A \hookrightarrow K_q$ for both primes. In particular A is a field and its degree is 4, 8, or 16. Degree eight would identify A with both unique octic subfields, contrary to (1.1e), and degree sixteen would identify the two full Frobenius fields, again contrary to (1.1e). Hence

$$\mathrm{End}^0_{\mathbf{Q}}(B_8) = E, \qquad \mathrm{End}_{\mathbf{Q}\text{-HS}}\, V = E. \tag{1.1}$$

In particular B_8 is absolutely simple. The Lefschetz isomorphism and the Rosati calculation above give

$$\mathrm{NS}(B_8)_{\mathbf{Q}} \simeq E^+, \qquad \rho(B_8) = 2. \tag{1.2}$$

For completeness, the four signatures can be read directly from the branch residues. Let $\langle b\rangle_{33} \in \{1, \dots, 32\}$ denote the nonzero residue of b. The four embeddings of E correspond to the $\langle 4\rangle$-orbits in $(\mathbf{Z}/33)^\times$. For a representative u, Chevalley–Weil gives

$$p_u = -1 + \frac{1}{33}\sum_{a\in\alpha}\langle ua\rangle_{33}, \qquad q_u = -1 + \frac{1}{33}\sum_{a\in\alpha}\langle -ua\rangle_{33}. \tag{1.2a}$$

Explicitly,

u	$\langle 4\rangle u$	$33^{-1}\sum_a\langle ua\rangle_{33}$	$33^{-1}\sum_a\langle -ua\rangle_{33}$
1	$\{1, 4, 16, 25, 31\}$	3	3
2	$\{2, 8, 17, 29, 32\}$	3	3
5	$\{5, 14, 20, 23, 26\}$	3	3
7	$\{7, 10, 13, 19, 28\}$	3	3

(1.2b)

The two residue sums are constant on each row because multiplication by (4) permutes the branch multiset α. The same Chevalley–Weil formula applied directly to C proves the source dimension independently of the Prym transfer. Its six branch exponents are the multiset α. For every unit $u \pmod{33}$, the two conjugate holomorphic character multiplicities are

$$m_u = -1 + \frac{1}{33}\sum_{a\in\alpha}\langle ua\rangle_{33} = 2,$$

$$m_{-u} = -1 + \frac{1}{33}\sum_{a\in\alpha}\langle -ua\rangle_{33} = 2.$$

Both equalities follow from (1.2b). Since $H^{0,1}(C)_{\chi_u}$ is conjugate to $H^{1,0}(C)_{\chi_{-u}}$, every primitive δ-eigenspace of total $H^1(C)$ has dimension $m_u + m_{-u} = 4$. Hence

$$\dim_K e_{33}H^1(C, \mathbf{Q}) = 4.$$

Consequently $U = \bigwedge^4_K e_{33}H^1(C, \mathbf{Q})$ is one-dimensional over K. The order-five action ρ is semilinear for K/E, so Galois descent makes U^ρ one-dimensional over E, hence four-dimensional over $\mathbf{Q}$. This establishes the source dimension asserted after the definition of U, without using the curve-to-Prym isomorphism. Thus $\dim_E V = 4$, and every embedding has Hodge signature $(p_u, q_u) = (2, 2)$. Hence V is an E-Hodge structure with Hodge numbers $(2, 2)$. By (1.1), its $\mathbf{Q}$-Hodge endomorphism algebra is the field E; semisimplicity of polarizable Hodge structures implies that V is simple. In Flapan's notation $n = 8$, $r = [E : \mathbf{Q}] = 4$, and the E-Hodge number $p = n/r = 2$ is prime. The centralizer is $C \simeq M_4(E^{\mathrm{op}})$, so all hypotheses of [17, Proposition 11.3] hold, giving

$$Hg(B_8) = \mathrm{Res}_{E^+/\mathbf{Q}}\, SU_E(V, h). \tag{1.3}$$

After extension to $\mathbf{C}$, this group is

$$Hg(B_8)_{\mathbf{C}} \simeq SL(W_1) \times SL(W_2), \qquad \dim W_1 = \dim W_2 = 4, \tag{1.4}$$

with

$$V_{\mathbf{C}} \simeq (W_1 \oplus W_1^\vee) \oplus W_2 \oplus W_2^\vee.$$

2. Main theorem and transfer

Conditional transfer theorem

For the fixed B_8 above, the following unconditional geometric assertions and conditional transfer statements are the current claims.

1. The endomorphism and Hodge-group identities are (1.1)–(1.3).
2. Subject to the source and finite-quotient inputs H1–H4 below, there is a finite extension

   $$K_0 = \mathbf{Q}(\zeta_{165}) \subset K' = K_0(\sqrt[3]{5}) \subset L$$

   and an embedding $L \hookrightarrow \mathbf{C}$ such that

   $$W_E(B_8) \subset \mathrm{Im}\Big(CH^2(B_{8,L})_{\mathbf{Q}} \xrightarrow{\mathrm{cl}} H^4(B_8, \mathbf{Q})\Big). \tag{2.1}$$

   Here K_0 defines the marked Kummer model, K' supplies the scalar needed for the common quotient, and L/K' is allowed to be a further finite extension for the selected source cycles, transfer correspondences, and the finite set of endomorphisms used to span the determinant line.
3. Subject to (2.1), for every $n \geq 1$ and $0 \leq p \leq 8n$,

   $$H^{2p}(B_8^n, \mathbf{Q}) \cap H^{p,p}(B_8^n) = \mathrm{Im}\Big(CH^p(B_{8,\mathbf{C}}^n)_{\mathbf{Q}} \xrightarrow{\mathrm{cl}} H^{2p}(B_8^n, \mathbf{Q})\Big). \tag{2.2}$$

   Thus the classical rational Hodge conjecture holds for every self-power.
4. Subject to (2.1), for every $n \geq 1$, the generalized Hodge conjecture holds for B_8^n.

Sections 3–6 prove the implication from H1–H4 to (2.1). In particular, the global small-diagonal correspondence in Section 6 bypasses the alternative Abel–Fourier and refined-Gysin route; no equality between those two Chow correspondences is asserted or needed. The all-powers classical Hodge statement (2.2) is conditional on (2.1) and is proved in Section 7 from the special-unitary description (1.3) and the rational determinant classes in (2.1). Section 8 applies Abdulali's Theorem 5.1 [10], with its later correction [18], separately to each self-power once (2.1) has been established.

Algebraic input

The source and target constructions used below are summarized by the following proposition; its proof is distributed through Sections 3–6. Items (1) and (2) are the content of H1–H4 together with the algebraicity of the source block; items (3) and (4) are proved unconditionally. The proposition is used only through the implication H1–H4 $\Rightarrow$ (2.1).

Proposition 2.1. Assume H1–H4. For the character $\alpha = (7, 10, 13, 19, 22, 28)$, the following statements hold after base change to the number field L of the Main Theorem:

1. The normalized quotient models satisfy $Y = C^4/G \simeq X_{33}^4/K_\alpha$, and the complete top-dimensional correspondence $\Gamma = [C^4 \times_Y X_{33}^4]_{\mathrm{top}}$ satisfies the finite-quotient pairing identity (4.2) and exact scaled isomorphism (4.3). Its restriction $\Gamma_*|_{U^\rho}$ has the explicit left inverse $(|G||K_\alpha|)^{-1}R_\Gamma$.
2. The exact-conductor Fermat block M_α is algebraic by the cubic-to-Fermat correspondence and rational unit-orbit projector in Section 6.2; the level-66 Aoki construction gives an independent proof. Also $\Gamma_* U^\rho \subseteq M_\alpha$. The adjoint $R_\Gamma = P_U^\rho \, {}^t\Gamma_* : M_\alpha \to U^\rho$ has the explicit right inverse $(|G||K_\alpha|)^{-1}\Gamma_*|_{U^\rho}$.

3. Put
$$H = (e_{33}H^1(C, \mathbf{Q}))^\rho.$$
There is a rational divisor correspondence
$$p \in CH^1(C \times B_8)_{\mathbf{Q}}$$
such that
$$p_*|_H : H \xrightarrow{\sim}_E H^1(B_8, \mathbf{Q}).$$
Under canonical Galois descent $\kappa_{\mathrm{desc}} : U^\rho \xrightarrow{\sim} \bigwedge^4_E H$, the actual small-diagonal transfer is
$$\bar{q}^{(4)}_* := 5^{-3} \bigwedge_E^4 (p_*|_H) \circ \kappa_{\mathrm{desc}} : U^\rho \xrightarrow{\sim} W_E(B_8).$$

4. If π_1^C is the degree-one Chow–Kunneth projector and $\Delta_B : B_8 \hookrightarrow B_8^4$ is the small diagonal, then
$$T_B := {}^t\Gamma_{\Delta_B} \circ (p \circ \pi_1^C)^{\boxtimes 4} \in CH^4(C^4 \times B_8)_{\mathbf{Q}}$$
sends every algebraic representative of U^ρ to an algebraic class spanning $W_E(B_8)$.

Proof. The valuation and common-coordinate calculation in the fixed model normalization subsection prove the quotient identification in (1); the exact finite-quotient formula is proved in Section 4. For (2), Section 6.2 proves M_α algebraic by mapping the full primitive cubic $(2,2)$-space onto its Fermat character block; Section 3 gives an independent proof via the level-66 lift of [28, Theorem 1.4]. Section 4 gives $\Gamma_* U^\rho \subseteq M_\alpha$, and Section 4 gives ${}^t\Gamma_*\Gamma_* = |G||K_\alpha|\,\mathrm{id}$ on $H^4(A, \mathbf{Q})^G$. For $u \in U^\rho$, this implies
$$R_\Gamma(\Gamma_* u) = |G||K_\alpha| u.$$
Thus $(|G||K_\alpha|)^{-1}\Gamma_*|_{U^\rho}$ is a right inverse to R_Γ. For (3), let $q_J^* : J(D) \to J(C)$ be the Jacobian pullback and put $\phi = q_J^* \circ i : B_8 \to J(C)$. Pulling the normalized Poincare divisor on $C \times J(C)$ back along $\mathrm{id}_C \times \phi$, and dividing by N, gives p. For the based Abel map $a_D : D \to J(D)$, put $\nu_D = (a_D^*)^{-1} : H^1(D, \mathbf{Q}) \to H^1(J(D), \mathbf{Q})$. Poincare naturality gives
$$p_*|_H = N^{-1} i^* \nu_D q_*|_H = (r^*)^{-1} \nu_D q_*|_H = \bar{q}_*.$$
The finite-quotient pull–push identities prove that this is an E-linear isomorphism. Galois descent and fourth exterior power give the abstract determinant isomorphism; the normalized synchronous orbit projector contributes the factor 5^{-3} in the actual small-diagonal map, as computed in Section 5.

For (4), $(p \circ \pi_1^C)^{\boxtimes 4}$ first isolates the $(1,1,1,1)$-Kunneth block, while the transpose of the small diagonal acts as Δ_B^*, hence as cup product. Its restriction to U^ρ is therefore
$$\mathrm{cup} \circ p_*^{\otimes 4} = 5^{-3} \bigwedge_E^4 (p_*|_H) \circ \kappa_{\mathrm{desc}} = \bar{q}^{(4)}_*.$$
All operations are rational Chow correspondences over L. □

Under H1–H4, after the fixed embedding $L \hookrightarrow \mathbf{C}$, every class in $W_E(B_8)$ is the cohomology class of a rational codimension-two algebraic cycle on $B_{8,\mathbf{C}}$. The subscript $\mathbf{Q}$ records Chow coefficients; it does not assert descent of the model or of these cycles to $\mathbf{Q}$.

The hypotheses H1–H4

The proof is the following conjunction:

All Chow groups in the source and target constructions below are taken after the base change to the fixed field L in the Main Theorem; the notation $CH^\bullet(-)_{\mathbf{Q}}$ refers to rational coefficients.

- **(H1–H2) Quotient models.** The normalized finite models are identified as $Y = C^4/G \simeq X_{33}^4/K_\alpha$. Formula (4.3) identifies the full invariant blocks by a scaled isomorphism; on U^ρ, formula (4.4) gives the explicit cohomological retraction $R_\Gamma\Gamma_* = (|G||K_\alpha|)\mathrm{id}$. No Chow-level composition identity is used.

- **(H3) Fermat source block.** For this transfer it suffices that

$$\mathrm{pr}_{H(-\alpha)}(I_X \otimes_{\mathbf{Q}} \mathbf{C}) \neq 0, \quad \text{i.e. condition (6.12) of Section 6.}$$

  Here $I_X = \mathrm{Im}(CH^2(X_{33}^4)_{\mathbf{Q}} \to H^4(X_{33}^4, \mathbf{Q}))$. Section 6.2 proves the stronger full-block statement by an explicit monomial correspondence from the smooth order-11 cubic Y; its rational unit-orbit projection identifies the entire primitive cubic $(2,2)$-space with a Fermat block. Aoki's Theorem 1–4(i),(ii), with the level-66 lift and descent of [28, Theorem 1.4], gives an independent proof of the same source statement. The target transfer itself still needs only the single-line condition.

- **(H4) Exact-conductor curve-to-factor correspondence.** Put $H = (e_{33}H^1(C, \mathbf{Q}))^{\rho}$. The fixed quotient $q : C \to D$, the inclusion $i : B_8 \hookrightarrow J(D)$, the Jacobian pullback $q_J^* : J(D) \to J(C)$, and the normalized Poincare divisor on $C \times J(C)$ give

$$p \in CH^1(C \times B_8)_{\mathbf{Q}}$$

  with

$$p_*|_H = \bar{q}_* : H \xrightarrow{\sim}_E H^1(B_8, \mathbf{Q}).$$

  The hypotheses end here. The following formula is not an assumption; it is proved in Proposition 2.2 and recorded for later use.

- **Small-diagonal tensor–cup formula (Proposition 2.2).** The correspondence

$$T_B = {}^t\Gamma_{\Delta_B} \circ (p \circ \pi_1^C)^{\boxtimes 4} \in CH^4(C^4 \times B_8)_{\mathbf{Q}}$$

  acts on the $(1,1,1,1)$-block as

$$\Delta_B^* \circ p_*^{\otimes 4} = \mathrm{cup} \circ p_*^{\otimes 4}.$$

  Its restriction to U^{ρ} is $5^{-3} \bigwedge_E^4 (p_*|_H) \circ \kappa_{\mathrm{desc}} = \bar{q}_*^{(4)}$. Since the source correspondence already projects exactly to U^{ρ}, no target determinant projector is required.

The scheme normalization and finite-quotient pairing are supplied in Sections 2 and 4. Sections 5–6 construct the curve-to-Prym and small-diagonal Chow correspondences. Under the minimal source-line condition (6.12), the adjoint transfer has nonzero image in U^{ρ}, and the small diagonal sends it to a nonzero class in $W_E(B_8)$; under the stronger full-block input it maps M_{α} onto all of U^{ρ}.

For the present character, Aoki's Theorem 1–4(i),(ii), together with the Lefschetz $(1,1)$ surface inputs, da Silva's Corollary 2.3(b) [19], and finite-quotient descent, proves the stronger full-block source statement $M_{\alpha} \subset I_X \otimes \mathbf{C}$. The main transfer does not require this full block; the construction below proves the minimal source-line condition (6.12) directly. Choose the distinguished final coordinate 33 in $\beta' = (2, 32, 65; 33)$ and $\gamma' = (8, 41, 50; 33)$. Their sum is zero modulo 66, and da Silva's $(r, s) = (2, 2)$ contraction gives $\beta' \# \gamma' = (2, 32, 65, 8, 41, 50) \sim S$, hence claim$(S)$. This is not an application of Aoki's $(2,2)$ product rule. Aoki's Theorem 1–4(i) with $(r, s) = (4, 2)$ gives claim$(S * Q)$; Theorem 1–4(ii), with the same pair and $a + \delta_0 = Q + S$, gives claim(a). The finite quotient $X_{66}^4 \to X_{33}^4$ then descends the resulting claim to the level-33 character α. This is an existence statement, not an explicit cycle equation.

Target-side calculation

Let

$$p_1 = p \circ \pi_1^C, \qquad P = p_1^{\boxtimes 4} \in CH^4(C^4 \times B_8^4)_{\mathbf{Q}},$$

and let $\Delta_B : B_8 \hookrightarrow B_8^4$ be the small diagonal. Since

$${}^t\Gamma_{\Delta_B} \in CH^{32}(B_8^4 \times B_8)_{\mathbf{Q}},$$

the composition

$$T_B = {}^t\Gamma_{\Delta_B} \circ P$$

has codimension

$$4 + 32 - \dim B_8^4 = 4$$

in $C^4 \times B_8$. It therefore gives

$$(T_B)_* : CH^2(C^4)_{\mathbf{Q}} \longrightarrow CH^2(B_8)_{\mathbf{Q}}.$$

Proposition 2.2 (global small-diagonal transfer). Let κ_{desc} be the canonical Galois-descent map from U^ρ to $\bigwedge_E^4 H$. The global Chow correspondence T_B has no boundary or excess term on the $(1,1,1,1)$ Kunneth block, and

$$(T_B)_*|_{U^\rho} = 5^{-3} \bigwedge_E^4 (p_*|_H) \circ \kappa_{\mathrm{desc}} \quad \text{in } H^4(B_8, \mathbf{Q}). \tag{2.3a}$$

Proof. $P = p_1^{\boxtimes 4}$ is defined on all of $C^4 \times B_8^4$, and ${}^t\Gamma_{\Delta_B}$ is the graph of a regular embedding, so their composition is a global Chow correspondence. On a pure tensor in $H^1(C)^{\otimes 4}$, Kunneth functoriality followed by diagonal pullback gives

$$\alpha_1 \otimes \cdots \otimes \alpha_4 \longmapsto p_*\alpha_1 \cup \cdots \cup p_*\alpha_4.$$

The Koszul signs of diagonal pullback cancel those in normalized alternation. The normalization is a special case of the following descent calculation. Let L'/F be cyclic of degree n, let σ generate its Galois group, let H be an F-vector space of dimension r, and put $M = L' \otimes_F H$ with its natural semilinear action. After splitting L'/F, write $e_{a,j}$ for the copy of an F-basis vector e_a in the j-th summand, and $\omega_j = e_{1,j} \wedge \cdots \wedge e_{r,j}$. The canonical descent map sends $\sum_j \omega_j$ to $e_1 \wedge_F \cdots \wedge_F e_r$. Thus, for $u_n = n^{-1} \sum_j \omega_j$ and $h_a = n^{-1} \sum_j e_{a,j}$,

$$\kappa_\Omega(u_n) = n^{-1}(e_1 \wedge_\Omega \cdots \wedge_\Omega e_r) = n^{r-1}(h_1 \wedge_\Omega \cdots \wedge_\Omega h_r),$$

where κ_Ω is the scalar extension of the descent map and the fixed vector h_a corresponds to $n^{-1}e_a$ under $(M \otimes_F \Omega)^\sigma \simeq H \otimes_F \Omega$. Apply this with $L' = K = \mathbf{Q}(\zeta_{33})$, $F = E$, $n = 5$, and $r = 4$. For one five-character orbit, take $v_{a,j} = \rho^j v_{a,0}$, $h_a = 5^{-1} \sum_j v_{a,j}$, and $\omega_j = \mathrm{Alt}_4(v_{1,j} \otimes \cdots \otimes v_{4,j})$. The source projector is the synchronous average $u = 5^{-1} \sum_j \omega_j$, so $\kappa_{\mathrm{desc}}(u) = 5^3(h_1 \wedge_E \cdots \wedge_E h_4)$. Since $p_*\rho = p_*$, the small-diagonal correspondence sends u to the same single-block cup product as $p_*h_1 \wedge \cdots \wedge p_*h_4$. This proves the factor 5^{-3} in (2.3a). □

On $H^1(C)^{\otimes 4}$, the external product P acts as $p_*^{\otimes 4}$, while the transpose graph of the small diagonal acts as Δ_B^*. Hence

$$(T_B)_* = \Delta_B^* \circ p_*^{\otimes 4} = \mathrm{cup} \circ p_*^{\otimes 4}. \tag{2.3}$$

The normalized alternation contributes no factorial. The order-five source projector, however, averages the five determinant blocks synchronously, whereas canonical descent uses four separately averaged H^1-vectors. Thus

$$(T_B)_*|_{U^\rho} = 5^{-3} \bigwedge_E^4 (p_*|_H) \circ \kappa_{\mathrm{desc}} = \bar{q}_*^{(4)} : U^\rho \xrightarrow{\sim} W_E(B_8). \tag{2.4}$$

The projector $(\pi_1^C)^{\boxtimes 4}$ eliminates every other Kunneth block before the target correspondence is applied. Consequently there are no target pollution spaces and no determinant projector is needed.

The non-free locus

Lemma 2.3 (finite-group support of the non-free locus). Let $\mathcal{B} \subset \mathbf{P}^1$ be the six branch values of $C \to \mathbf{P}^1$, and let $U_{\mathrm{fr}} \subset \mathrm{Sym}^4(\mathbf{P}^1)$ be the locus of reduced degree-four divisors disjoint from $\mathcal{B}$. For

$$G = N \rtimes \mathfrak{S}_4, \qquad N = \{(t_1, \dots, t_4) \in (\mathbf{Z}/33)^4 : \textstyle\sum t_i = 0\},$$

every codimension-one component of the pullback of $\mathrm{Sym}^4(\mathbf{P}^1) \setminus U_{\mathrm{fr}}$ to C^4 is of one of the following two types:

$$\{x_j = p\} \quad (p \in C \text{ over } \mathcal{B}), \qquad \{x_j = \delta^r x_k\} \quad (r \in \mathbf{Z}/33). \tag{2.5a}$$

Proof. Over U_{fr}, the ordered pullback is an étale $(\mathbf{Z}/33)^4$-torsor and the four base points are distinct. If an element of G stabilizes a point over this locus, its permutation part must preserve four distinct base points and is therefore trivial; its deck part then fixes four unramified points, hence is also trivial. Thus a codimension-one component outside the free locus must either lie over a branch value or over a diagonal of $(\mathbf{P}^1)^4$. A branch component pulls back to $\{x_j = p\}$, with

the non-uniform inertia at exponent 22 only splitting this same type. A diagonal $z_j = z_k$ pulls back, on each deck sheet, to $x_j = \delta^r x_k$. Intersections of these divisors have codimension at least two and introduce no new generic component. □

Corollary 4.2 (opposite-character boundary criterion). The preceding support argument is uniform in the conductor. Let $N \geq 2$, let $q : C \to \mathbf{P}^1$ be a cyclic μ_N-cover, and suppose an r-fold common quotient, $r \geq 1$, is formed from

$$N_0 = \ker\big((\mathbf{Z}/N)^r \longrightarrow \mathbf{Z}/N\big) \rtimes \mathfrak{S}_r.$$

The codimension-one complement of the reduced, branch-avoiding locus in $\mathrm{Sym}^r(\mathbf{P}^1)$ consists of branch-point and pairwise-diagonal strata; away from it the action is free because a stabilizing permutation preserves distinct points and a deck element fixes no unramified point. Over $\mathbf{C}$, write $V_a = \{v \in H^1(C, \mathbf{C}) : \delta^* v = \zeta_N^a v\}$. For $\mathbf{n} = (n_1, \dots, n_r)$, put $V_{\mathbf{n}} = V_{n_1} \otimes \cdots \otimes V_{n_r}$. If $\mathcal{A} \subset (\mathbf{Z}/N)^r$ and every $\mathbf{n} \in \mathcal{A}$ satisfies $n_j + n_k \not\equiv 0 \pmod N$ for all $j \neq k$, then every codimension-one non-free boundary restriction vanishes on $V_{\mathcal{A}} = \bigoplus_{\mathbf{n} \in \mathcal{A}} V_{\mathbf{n}}$. Indeed, a branch factor restricts one $H^1(C)$-entry to a point. Along a twisted diagonal identifying factors j, k, the restriction factors through

$$V_{n_j} \otimes V_{n_k} \xrightarrow{\cup} H^2(C, \mathbf{C}),$$

whose source character is $n_j + n_k$, while $H^2(C, \mathbf{C})$ is deck-trivial. Thus only opposite-character pairs can contribute. Equivariant Poincare duality makes $V_a \otimes V_{-a} \to H^2(C, \mathbf{C})$ a perfect pairing whenever these eigenspaces are nonzero, so the criterion is sharp for the diagonal restriction map.

In particular, for the equal-character block with character n, let P_n be its projector. If $2n \not\equiv 0 \pmod N$, every codimension-one non-free boundary correspondence R satisfies

$$P_n R P_n = 0. \tag{2.5b}$$

Hence every primitive equal-character block is covered for every $N > 2$, including even conductors. The order-two exception is real: on a connected genus-one double cover, the anti-invariant H^1 has nonzero alternating cup pairing, so the diagonal restriction need not vanish. The criterion is independent of branch inertia and r; higher-codimension intersections do not contribute to the top-dimensional finite correspondence.

Let $Y^\circ \subset \mathrm{Sym}^4(\mathbf{P}^1)$ be the locus of four distinct points avoiding the six branch values of the Kummer cover $C \to \mathbf{P}^1$. The complement of the corresponding free-layer correspondence has, on the source side, only the following two types of irreducible support:

$$\{x_j = p\} \quad (p \text{ a branch point}), \qquad \{x_j = \delta^r x_k\} \quad (r \in \mathbf{Z}/33). \tag{2.5}$$

The first type is a point-factor; the second is a twisted diagonal. This follows because the only ways for an unordered four-point divisor to leave Y° are a branch point or a repeated point, and the finite deck group lifts equality of two quotient points to $x_j = \delta^r x_k$. Intersections of these loci give no new irreducible type. The non-uniform inertia at the exponent 22 branch merely splits the first type into $\gcd(33, 22) = 11$ point-factors.

The exact-conductor source projector kills both types. On a point-factor, one $H^1(C)$ factor restricts to zero. On a twisted diagonal, the two identified primitive character factors have cup character $2n$, while $H^2(C)$ carries only the trivial deck character; since $2n \not\equiv 0 \pmod{33}$ for every primitive n, the restriction is zero. By localization and the projection formula, any correspondence supported over the boundary therefore satisfies

$$P_U^\rho R P_U^\rho = 0 \quad \text{on } U^\rho. \tag{2.6}$$

Thus the order-three inertia at $\alpha_4 = 22$ creates no additional top-dimensional Chow correction. The support classification in (2.5), together with the character calculation in (2.6), gives the required boundary vanishing.

The lemma is a geometric support statement for the alternative Abel–Fourier presentation. The small-diagonal proof below uses the finite-quotient pairing instead and does not require a source Chow composition. The target transfer itself is the globally defined small-diagonal correspondence of Section 6. The projector and refined-Gysin scalar calculation belongs to the alternative Abel–Fourier presentation: for $\pi_B = N^{-1} ir$, the transpose graph acts as π_B^*, and $i^*(\pi_B^*)^{\wedge 4} = N^4 \,\mathrm{id}$ on $\bigwedge_E^4 r^* H^1(B_8, \mathbf{Q})$. This typed calculation is formal and is not an additional step in the small-diagonal correspondence (6.6).

Normalization of the model

The common quotient identification reduces to one concrete normalization test. Let

$$S = \{\sum_i Z_i' = 0\} \simeq \mathbf{P}^4, \qquad f = \prod_{i=0}^{5} (Z_i'/\ell)^{\alpha_i} \in \mathbf{C}(S)^\times,$$

where ℓ is chosen so that it does not vanish identically on the divisor $Z_0' = 0$. At that divisor,

$$v_{Z_0'}(f) = \alpha_0 = 7. \tag{2.10}$$

Here is the coordinate normalization that fixes the character label. On the dense affine chart of $\mathrm{Sym}^4(\mathbf{P}^1)$, write the generic divisor as $\{z_1, \dots, z_4\}$ and its binary quartic as $P(s,t) = \lambda_0 \prod_{j=1}^{4} (z_j s - t)$. Put

$$L_i = P(1, \zeta_5^i)\ (0 \le i < 5), \qquad L_\infty = P(0,1), \qquad Z_i = \zeta_5^i L_i, \quad Z_5 = -5L_\infty,$$

and order the six base coordinates by

$$(Z_0', \dots, Z_5') = (Z_0, Z_4, Z_2, Z_3, Z_5, Z_1).$$

Write $P(1,t) = \sum_{k=0}^{4} c_k t^k$; then $c_4 = P(0,1) = L_\infty$, and the root-of-unity filter gives

$$\sum_{i=0}^{4} \zeta_5^i L_i = \sum_{k=0}^{4} c_k \sum_{i=0}^{4} \zeta_5^{i(k+1)} = 5c_4 = 5L_\infty.$$

The five evaluations L_i are independent coordinates by the Vandermonde determinant, and L_∞ is their linear combination above. Thus the ordered coordinates Z_i' identify S with the hyperplane $\sum_i Z_i' = 0$. Their Kummer exponent tuple is

$$(a_0, a_4, a_2, a_3, -\textstyle\sum_{i=0}^{4} a_i, a_1) \equiv (7, 10, 13, 19, 22, 28) = \alpha \pmod{33}.$$

If $V = (\prod_{j=1}^{4} y_j) L_\infty^3$, then the homogeneous Kummer generator $w_C = V/\ell^3$ satisfies

$$w_C^{33} = \kappa f, \qquad \kappa = \zeta_5^4\, 5^{-22}.$$

Indeed, $L_\infty = \lambda_0$, so the affine Kummer equations give

$$\prod_{j=1}^{4} y_j^{33} = \prod_{i=0}^{4} (L_i/L_\infty)^{a_i}, \qquad (a_0, \dots, a_4) = (7, 28, 13, 19, 10).$$

Since $\sum_{i=0}^{4} a_i = 77$, it follows that

$$V^{33} = \prod_{i=0}^{4} L_i^{a_i} L_\infty^{22}.$$

In the displayed coordinate order,

$$\prod_{i=0}^{5} (Z_i')^{\alpha_i} = \zeta_5^{151} (-5)^{22} \prod_{i=0}^{4} L_i^{a_i} L_\infty^{22} = \zeta_5\, 5^{22} V^{33},$$

because $151 \equiv 1 \pmod 5$. Dividing by ℓ^{99} gives the stated $\kappa = \zeta_5^{-1} 5^{-22} = \zeta_5^4\, 5^{-22}$. If $r^3 = 5$, then

$$\lambda = \zeta_5^3 r^{-2} \in K' = \mathbf{Q}(\zeta_{165}, r), \qquad \lambda^{33} = \zeta_5^{99} r^{-66} = \zeta_5^4\, 5^{-22} = \kappa.$$

On the Fermat side the generator

$$w_F = \frac{\prod_{i=0}^{5} x_i^{\alpha_i}}{\ell (x_0^{33}, \dots, x_5^{33})^3}$$

satisfies $w_F^{33} = f$. Thus the chosen common-model isomorphism sends $w_C \mapsto \lambda w_F$ for any fixed λ with $\lambda^{33} = \kappa$; the residual generators act with the same multiplier. Since 7 is prime to 33, the Kummer binomial $T^{33} - f$ is irreducible

over $\mathbf{C}(S)$: f is neither a cube nor an eleventh power. The Kummer quotient $C^4/G \to S$ and the Fermat character quotient $X_{33}^4/K_\alpha \to S$ are finite normal models with this same function-field extension. More explicitly, $C^4 \to (\mathbf{P}^1)^4$ is finite of degree 33^4; quotienting by N, of order 33^3 and trivial on the base, gives a finite degree-33 map, which descends through $\mathfrak{S}_4$ to $C^4/G \to \mathrm{Sym}^4(\mathbf{P}^1) \simeq S$. On the Fermat side, the coordinatewise 33-rd-power map restricts to a finite degree-33^5 map $X_{33}^4 \to S$; quotienting by K_α, of order 33^4, gives a finite degree-33 map. Both quotient varieties are normal as finite-group quotients of smooth varieties in characteristic zero. Their common Kummer generator T, after adjoining the stated scalar root, identifies the degree-33 function-field extensions. Thus both are the normalization of S in the same extension, and normalization uniqueness gives the scheme-level identification in (H1).

The coordinate permutation, valuation (2.10), Kummer degree, and group orders are checked by the accompanying arithmetic calculations. This is a fixed model normalization argument; it makes no claim about Chow-composition identities for arbitrary birational models.

The extension $K' = K_0(\sqrt[3]{5})$ is minimal for an isomorphism over this fixed base S. Indeed, let E/K_0 be any extension and suppose the two degree-33 function-field extensions are $E(S)$-isomorphic. The image of w_C is an eigenvector for the deck generator $w_F \mapsto \zeta_{33} w_F$, hence has the form $h w_F^j$, with $h \in E(S)$ and $j \in (\mathbf{Z}/33)^\times$. Comparing 33rd powers and taking the valuation at $Z_0' = 0$ gives

$$33 v_{Z_0'}(h) + 7j = 7,$$

so $j = 1 \pmod{33}$, and then $h^{33} = \kappa$. Since $E(S)/E$ is regular, this implies $\kappa \in E^{\times 33}$. Conversely, any 33rd root of κ in E gives the required isomorphism. Every such root is $\lambda = \zeta_5^3(\sqrt[3]{5})^{-2}\xi$ with $\xi \in \mu_{33} \subset K_0$; therefore $\sqrt[3]{5} = 5\lambda/(\zeta_5^3\xi) \in E$. Thus the isomorphism exists over $E(S)$ exactly when $K' \subseteq E$. In particular, $[K' : K_0] = 3$: $T^3 - 5$ is irreducible over K_0: since $\mu_3 \subset K_0$, a root would put $\mathbf{Q}(\sqrt[3]{5})$ inside the abelian extension $K_0/\mathbf{Q}$, contradicting that $\mathbf{Q}(\sqrt[3]{5})/\mathbf{Q}$ is not normal. Equivalently, the class of κ in $K_0^\times/(K_0^\times)^{33}$ has exact order three: ζ_5^4 is a 33rd power in K_0, $\kappa^3 = (\zeta_5^4)^3(5^{-2})^{33}$, and $v_{\mathfrak{p}}(\kappa) = -88 \notin 33\mathbf{Z}$. This minimality is relative to the fixed coordinates on S; it makes no claim for isomorphisms covering a nontrivial automorphism of S, or for the field of definition of the source cycles and the full transfer.

3. Fermat source via the level-66 lift

The Hodge rings of abelian varieties of Fermat type, and the question of when their codimension ≥ 2 classes are generated by divisors, were studied systematically by Shioda [14]; the source computation below takes place in that setting and uses no claim of originality about the transfer framework itself.

Put

$$w = (1, 4, 16, 22, 25, 31), \qquad a = 2w \pmod{66}.$$

The balanced Hodge type of the target character can be checked without enumerating all twenty units. Let

$$H = \langle 4 \rangle = \{1, 4, 16, 25, 31\} \subset (\mathbf{Z}/33)^\times.$$

The four cosets of H have representatives $1, 2, 5, 7$; moreover $22h \equiv 22 \pmod{33}$ for every $h \in H$. Their least-positive residues give

r	rH (in increasing order)	$\sum rH$	$[22r]_{33}$	$\sum rH + [22r]_{33}$
1	$(1, 4, 16, 25, 31)$	77	22	99
2	$(2, 8, 17, 29, 32)$	88	11	99
5	$(5, 14, 20, 23, 26)$	88	11	99
7	$(7, 10, 13, 19, 28)$	77	22	99

Thus $\sum_i [uw_i]_{33} = 99$ for every unit u. Since $\alpha = 7w \pmod{33}$, this proves $\sum_i [u\alpha_i]_{33} = 99 = 3 \cdot 33$ for every unit u, so the full rational orbit M_α has Fermat Hodge type $(2, 2)$. For a unit $t \pmod{66}$, its reduction $u \pmod{33}$ satisfies $[ta_i]_{66} = 2[uw_i]_{33}$; hence $\sum_i [ta_i]_{66} = 198 = 3 \cdot 66$, verifying the same condition for a on X_{66}^4. The separate unit-Hodge checks for Q, S, β, γ are unchanged.

The exact finite identity

$$a + \delta_0 = Q + S$$

with

$$Q = (1, 25, 44, 62), \quad S = (2, 8, 32, 41, 50, 65),$$
$$\delta_0 = (1, 65, 25, 41), \quad \beta = (2, 32, 33, 65),$$
$$\gamma = (8, 33, 41, 50)$$

hold as multiset identities at level 66, and all unit Hodge equalities hold. For a Fermat character b, write claim(b) for the assertion that its rational Galois-orbit Hodge block is generated by algebraic cycles, and write $*$ for Aoki's join operation. The four-tuples Q, β, γ are the grade-two (Fermat-surface) characters in these identities. If V_b denotes the rational Galois-orbit block of the character b in the corresponding Fermat surface factor, then the unit Hodge equalities put

$$V_Q,\ V_\beta,\ V_\gamma \subset H^{1,1}(\Sigma_{66}) \cap H^2(\Sigma_{66}, \mathbf{Q}),$$

where Σ_{66} is the level-66 Fermat surface used in Aoki's construction. The Lefschetz $(1,1)$ theorem therefore gives

$$V_Q,\ V_\beta,\ V_\gamma \subset \mathrm{NS}(\Sigma_{66})_{\mathbf{Q}}. \tag{3.0}$$

After base change to $\mathbf{C}$, these Néron–Severi classes are the base changes of classes on $\Sigma_{66,\overline{\mathbf{Q}}}$; a chosen divisor representative is consequently defined over a finite extension of the cyclotomic field of definition. Thus

$$\text{claim}(Q), \qquad \text{claim}(\beta), \qquad \text{claim}(\gamma)$$

are algebraic surface inputs. Aoki's theorem is used below only for the subsequent product and final elimination steps. For $\beta' = (2, 32, 65; 33)$ and $\gamma' = (8, 41, 50; 33)$, da Silva's Corollary 2.3(b) [19] applies with $(r, s) = (2, 2)$: the distinguished final entries sum to zero modulo 66, and deleting them gives S, up to permutation. Thus claim(S). This contraction is distinct from Aoki's juxtaposition operation. Aoki's statement of the Shioda–Ran theorem [20, Theorem 1–4, p. 388] applies with $(r, s) = (4, 2)$. Part (i), with $(\alpha, \beta) = (S, Q)$, uses both established inputs claim(S) and claim(Q) to give claim($S*Q$); part (ii), with $(\alpha, \delta) = (a, \delta_0)$, gives claim($a$) because $a*\delta_0 = S*Q$ up to permutation and $\delta_0 \in \mathfrak{D}^2_{66}$. The unit-sum tests give $S, a \in \mathfrak{B}^4_{66}$ and $Q \in \mathfrak{B}^2_{66}$, as required by the theorem. Here $\delta_0 = (1, -1, 25, -25) \pmod{66}$ is a Shioda opposite-pair character, so $\delta_0 \in \mathfrak{D}^2_{66}$, the additional hypothesis in part (ii). Both applications have Aoki's dimension index $n = r + s + 2 = 8$; the joined character has length $n + 2 = 10$, as required for the eight-dimensional Fermat variety. Part (ii) contracts the $\mathfrak{D}^2_{66}$-factor and returns the fourfold character a. Hence the level-66 fourfold character a is generated by algebraic cycles in Aoki's sense. Aoki's theorem is stated for a single complex eigenspace, whereas our claim denotes the full rational Galois-orbit block. To pass between them, apply the same argument after multiplying every displayed tuple by each $t \in (\mathbf{Z}/66)^\times$. This preserves the unit-sum and multiset identities, preserves the opposite pairs in δ_0, and fixes the distinguished entry 33 modulo 66; hence the Lefschetz, da Silva, and Aoki steps give algebraicity of every eigenspace $V(ta)$. The distinct such eigenspaces form $M_{66}(a) \otimes \mathbf{C}$, so the orbit-level claim follows.

For a projective variety X/K_0, the images of $CH^p(X_{\overline{K_0}})_{\mathbf{Q}}$ and $CH^p(X_{\mathbf{C}})_{\mathbf{Q}}$ in $H^{2p}(X_{\mathbf{C}}, \mathbf{Q})$ coincide. Indeed, reduce to an integral component Z of a complex cycle and its point on the Chow variety of cycles of the same dimension and degree. The geometric connected component containing that point descends to $\overline{K_0}$, has an $\overline{K_0}$-point, and the universal cycle class is constant on the component. Thus Z has a $\overline{K_0}$-defined cycle representative with the same class. Applying this to the finitely many eigenspace cycles supplied by Aoki gives representatives over one finite extension of K_0: each point on a finite-type Chow variety is defined over a finite extension, and only finitely many unit conjugates occur. This justifies the fixed field L.

The field L cannot be taken to be $\mathbf{Q}(\zeta_{33})$: by Jumagulov [28, Theorem 1.6], no codimension-two cycle defined over $\mathbf{Q}(\zeta_{33})$ has nonzero projection onto the block M_α, and by [28, Corollary 1.7] every finite field of definition of such a cycle extending $\mathbf{Q}(\zeta_{33})$ has all residue degrees above 67 divisible by 6. The arguments here use only that L is some finite extension of the cyclotomic field. The factor 5^{-3} in (2.3a) concerns descent of coefficients from $K = \mathbf{Q}(\zeta_{33})$ to E on cohomology; it is not a descent of cycles to a smaller field of definition.

The finite quotient

$$\pi : X^4_{66} \longrightarrow X^4_{33}, \qquad [x_0 : \cdots : x_5] \longmapsto [x_0^2 : \cdots : x_5^2]$$

has group $\mu_2^6/\mu_{2,\mathrm{diag}}$ of order 32. Write $M_m(b)$ for the rational Galois-orbit block of character b on X^4_m. The diagonal character w pulls back to $2w$: for $\xi = (\xi_i) \in \mu_{66}^6/\mu_{66,\mathrm{diag}}$, $\chi_w(\xi_0^2, \ldots, \xi_5^2) = \chi_{2w}(\xi)$. Hence $\pi^* : M_{33}(w) \to M_{66}(2w)$.

Since $a = 2w$ is coordinatewise even, every element of the kernel acts trivially on $M_{66}(2w)$. For this finite quotient, the rational cohomological correspondence identities are

$$\pi^*\pi_* = \sum_{g \in \mu_2^6/\mu_{2,\mathrm{diag}}} g^*, \qquad \pi_*\pi^* = 32\ \mathrm{id}\,.$$

Restricting the first identity to the kernel-invariant block gives

$$\pi^*\pi_*|_{M_{66}(2w)} = 32\ \mathrm{id}, \qquad \pi_*\pi^*|_{M_{33}(w)} = 32\ \mathrm{id}\,.$$

Equivariance also gives $\pi_* M_{66}(2w) \subseteq M_{33}(w)$. Thus $\pi^* : M_{33}(w) \xrightarrow{\sim} M_{66}(2w)$ and $\pi_*|_{M_{66}(2w)} = 32(\pi^*)^{-1}$. Pushing forward Aoki's algebraic cycles and dividing by 32 therefore gives algebraic cycles spanning $M_{33}(w)$. Since $7w$ is α up to a coordinate permutation, the exact-conductor block $M_\alpha \subset H^4(X_{33}^4, \mathbf{Q})$ is generated by algebraic cycles. Let

$$A_{33} = \mathrm{Im}\left(CH^2(X_{33}^4)_{\mathbf{Q}} \xrightarrow{\mathrm{cl}_{\mathbf{Q}}^2} H^4(X_{33}^4, \mathbf{Q}) \right).$$

The preceding all-conjugates argument and finite pushforward give $M_\alpha \otimes_{\mathbf{Q}} \mathbf{C} \subseteq A_{33} \otimes_{\mathbf{Q}} \mathbf{C}$. Faithful extension of scalars from $\mathbf{Q}$ to $\mathbf{C}$ then gives $M_\alpha \subseteq A_{33}$. Equivalently,

$$M_\alpha \subseteq \mathrm{Im}\left(CH^2(X_{33}^4)_{\mathbf{Q}} \xrightarrow{\mathrm{cl}_{\mathbf{Q}}^2} H^4(X_{33}^4, \mathbf{Q}) \right). \tag{3.1}$$

For the determinant transfer, the single source eigenspace needed in Section 6.2 is already visible before completing the full orbit. The unit $59 \pmod{66}$ gives

$$59a \equiv -14w = \pi^*(-7w) \pmod{66}, \qquad -7w \equiv -\alpha \pmod{33}$$

up to coordinate permutation. Multiplying every tuple in the displayed Aoki–da Silva identities by 59 preserves the unit-sum and multiset identities; it sends δ_0 to $(59, 7, 23, 43)$, with opposite pairs $(59, 7)$ and $(23, 43)$, and fixes the distinguished residue 33. The same single-eigenspace argument therefore gives an algebraic cycle class on X_{66}^4 with nonzero $59a$-component. Write $c = -7w \pmod{33}$, so $59a = 2c \pmod{66}$. The quotient-character relation puts this component in the c-eigenspace after pushforward. Every entry of $59a$ is even, so the order-32 kernel K of π acts trivially on the component $\eta \neq 0$. Hence

$$\pi^*\pi_*\eta = \sum_{g \in K} g^*\eta = 32\eta \neq 0,$$

and therefore $\pi_*\eta \neq 0$. This is an algebraic class in $H(c)$. Since $c = -7w$ is a coordinate permutation of $-\alpha$, the corresponding coordinate-permutation automorphism gives a nonzero algebraic projection to $H(-\alpha)$. Thus the source condition (6.12) follows from this one conjugate; the preceding all-conjugates argument proves the stronger full-block statement (3.1).

4. Common quotient and adjoint transfer

Let

$$G = N \rtimes \mathfrak{S}_4, \qquad N = \{(t_1, \dots, t_4) \in (\mathbf{Z}/33)^4 : \textstyle\sum t_i = 0\},$$

and let $K_\alpha = \ker \chi_\alpha$. The Kummer invariant-field calculation and normalization uniqueness give the same finite normal model

$$Y = C^4/G \simeq X_{33}^4/K_\alpha. \tag{4.1}$$

Let

$$\Gamma = [C^4 \times_Y X_{33}^4]_{\mathrm{top}}.$$

Write $A = C^4$, $X = X_{33}^4$, and let $q_A : A \to Y$ and $q_X : X \to Y$ be the finite quotient maps. Put $W = A \times_Y X$, with projections $\pi_A : W \to A$ and $\pi_X : W \to X$. Both are finite, and $\Gamma = [W]_{\mathrm{top}}$.

Indeed, on the dense open of X where all six homogeneous coordinates are nonzero, a diagonal projective automorphism fixing a point has all six multipliers equal, hence is the identity in the effective projective diagonal group. Thus the

finite K_α-action is free there, and q_X is a K_α-torsor over a dense open $Y^\circ \subset Y$. Its base change to $A^\circ = q_A^{-1}(Y^\circ)$ is finite étale of degree $|K_\alpha|$. Thus the generic rank of π_A is $|K_\alpha|$, which is the coefficient of $[A]$ in the pushforward of the top fundamental cycle, including generic scheme lengths. The inverse image of $A \setminus A^\circ$ has dimension at most three because π_A is finite, so it contributes no four-dimensional component. Hence

$$(\pi_A)_*[W]_{\mathrm{top}} = |K_\alpha|[A].$$

The action of G on A is faithful: a nontrivial permutation changes the product factors, and with trivial permutation a nonzero deck exponent acts nontrivially on its curve factor. A faithful finite group action in characteristic zero is free on a dense open, so q_A has generic degree $|G|$. Since π_X is its base change along q_X,

$$(\pi_X)_*[W]_{\mathrm{top}} = |G|[X].$$

Lemma 4.1 (finite-quotient pairing). For $u \in H^4(A, \mathbf{Q})^G$ and $v \in H^4(Y, \mathbf{Q})$,

$$\langle \Gamma_* u, q_X^* v \rangle_X = |K_\alpha| \, \langle u, q_A^* v \rangle_A. \tag{4.2}$$

In fact, q_A^* and q_X^* identify $H^4(Y, \mathbf{Q})$ with the respective invariant cohomology, and

$$\Gamma_*|_{H^4(A,\mathbf{Q})^G} = |G| \, q_X^*(q_A^*)^{-1} : H^4(A, \mathbf{Q})^G \xrightarrow{\sim} H^4(X, \mathbf{Q})^{K_\alpha}. \tag{4.3}$$

Thus the restriction is a scaled isomorphism, not merely injective.

Proof. For either finite quotient $q : Z \to Y$ by a group H, proper base change gives $R^i q_* \mathbf{Q}_Z = 0$ for $i > 0$, and $(q_* \mathbf{Q}_Z)^H = \mathbf{Q}_Y$, since each fiber is one H-orbit. Exactness of invariants over $\mathbf{Q}$ and Leray therefore identify $q^* : H^4(Y, \mathbf{Q}) \xrightarrow{\sim} H^4(Z, \mathbf{Q})^H$. Let $[W]_{\mathrm{BM}}$ denote the Borel–Moore class of the top cycle, with its generic scheme lengths. Cycle-class compatibility with cap products and proper pushforward gives, for $u \in H^4(A, \mathbf{Q})$,

$$\mathrm{PD}_X(\Gamma_* u) = (\pi_X)_*(\pi_A^* u \cap [W]_{\mathrm{BM}}).$$

Thus for $a \in H^4(Y, \mathbf{Q})$, using $q_A \pi_A = q_X \pi_X$ and the two top-cycle pushforwards,

$$\begin{aligned} \mathrm{PD}_X(\Gamma_* q_A^* a) &= (\pi_X)_*(\pi_X^* q_X^* a \cap [W]_{\mathrm{BM}}) \\ &= q_X^* a \cap (\pi_X)_*[W]_{\mathrm{BM}} = |G| \, \mathrm{PD}_X(q_X^* a), \\ \mathrm{PD}_A({}^t\Gamma_* q_X^* a) &= (\pi_A)_*(\pi_A^* q_A^* a \cap [W]_{\mathrm{BM}}) \\ &= q_A^* a \cap (\pi_A)_*[W]_{\mathrm{BM}} = |K_\alpha| \, \mathrm{PD}_A(q_A^* a). \end{aligned}$$

This uses no Chow pullback to the possibly singular W. It follows that

$$\Gamma_* q_A^* a = |G| q_X^* a, \qquad {}^t\Gamma_* q_X^* a = |K_\alpha| q_A^* a.$$

The first equality proves (4.3). The second and correspondence adjunction give (4.2). Composing the two pull–push formulas also gives

$${}^t\Gamma_* \circ \Gamma_*|_{H^4(A,\mathbf{Q})^G} = |G||K_\alpha| \, \mathrm{id}_{H^4(A,\mathbf{Q})^G}. \tag{4.4}$$

This is an identity on cohomology, not an equality of Chow correspondence cycles. □

More explicitly, on cohomology the source projector is

$$P_U^\rho = p_\rho p_G e_{33}^{\boxtimes 4}, \qquad p_\rho = \frac{1}{5} \sum_{j=0}^{4} \Gamma_{(\rho^j)\times 4}.$$

The factors commute because the diagonal ρ-action normalizes G and preserves the exact-conductor block. Each is self-adjoint: the group averages are stable under inversion, and the exact-conductor projector is stable under transposition. Hence P_U^ρ is a self-adjoint projector and the Poincare pairing on U^ρ is nondegenerate. To check its image, extend scalars to $\overline{\mathbf{Q}}$. The projector e_{33} kills $H^0(C)$ and $H^2(C)$, so the image lies in $V^{\otimes 4}$. Decompose $V = \bigoplus_{k \in (\mathbf{Z}/33)^\times} V_k$ into δ-eigenspaces, with δ acting on V_k by ζ_{33}^k. On $V_{k_1} \otimes \cdots \otimes V_{k_4}$, the subgroup N acts by $(t_i) \mapsto \zeta_{33}^{\sum_i t_i k_i}$. Its annihilator

is the diagonal: testing $t = e_i - e_j$ forces $k_i = k_j$, and equal k_i annihilate N. Thus the N-invariants are $\bigoplus_k V_k^{\otimes 4}$. Since $p_G = p_{\mathfrak{S}_4} p_N$, it remains to take the geometric $\mathfrak{S}_4$-invariants on these blocks. Write τ_σ for the ungraded permutation of tensor factors. Because each factor has degree one, the geometric action is $\mathrm{sgn}(\sigma)\tau_\sigma$; hence

$$p_{\mathfrak{S}_4}|_{V_k^{\otimes 4}} = \frac{1}{24} \sum_{\sigma \in \mathfrak{S}_4} \mathrm{sgn}(\sigma)\tau_\sigma = \mathrm{Alt}_4 .$$

Its image is the ordinary alternating tensor space $\bigwedge^4 V_k$. Therefore $\mathrm{Im}(p_G e_{33}^{\boxtimes 4}) = (\bigwedge_K^4 V)_{\overline{\mathbf{Q}}} = U_{\overline{\mathbf{Q}}}$. The diagonal ρ-action normalizes G and preserves $e_{33}^{\boxtimes 4}$, hence p_ρ projects this block onto U^ρ. In particular, $P_U^\rho p_G = P_U^\rho$.

It remains to check that $\Gamma_* U^\rho$ lies in the block covered by (3.1). On the curve-product side, put $\Delta_C = (\mu_{33})^4$. Its residual quotient $\Delta_C/N \simeq \mu_{33}$ is generated by the class of $(\zeta_{33}, 1, 1, 1)$; this class normalizes G, and acts on the Kummer coordinate $V = (\prod_i y_i)L_\infty^3$ of C^4/G by $V \mapsto \zeta_{33} V$. On the Fermat side, let $A = (\mu_{33})^6/\mu_{33,\mathrm{diag}}$, so that $A/K_\alpha \simeq \mu_{33}$ via χ_α. The element $a = [(\zeta_{33}^{19}, 1, 1, 1, 1, 1)]$ satisfies $\chi_\alpha(a) = \zeta_{33}$, since $7 \cdot 19 \equiv 1 \pmod{33}$. It multiplies the Fermat Kummer generator $w_F = (\prod_i x_i^{\alpha_i})/\ell_X^3$ by ζ_{33}, where $\ell_X = \ell(x_0^{33}, \dots, x_5^{33})$. On the common base of (4.1), put $w_C = V/\ell^3$ and $f = \prod_i (Z_i'/\ell)^{\alpha_i}$. The fixed coordinate ordering in Section 2 gives $w_C^{33} = \kappa f$ and $w_F^{33} = f$, for some $\kappa \in \mathbf{C}^\times$. The normalized isomorphism sends $w_C \mapsto \lambda w_F$, where $\lambda^{33} = \kappa$. Hence the two residual automorphisms agree; the scalar does not alter their multiplier and the fixed coordinate ordering α introduces no inverse relabelling. To record the direction of equivariance, write $\tilde{\delta}_C = (\zeta_{33}, 1, 1, 1)$ for the automorphism of C^4 representing the residual generator, and write a_X for the chosen automorphism of X_{33}^4. Write $\delta^* = \tilde{\delta}_C^*$ on cohomology, as in the definition of V_n below. They induce the same automorphism of Y: both fix the common base and multiply the normalized Kummer generator by ζ_{33}. Thus

$$g_W : W \longrightarrow W, \qquad (c, x) \longmapsto (\tilde{\delta}_C c, a_X x)$$

is well defined. In particular, $h = \tilde{\delta}_C \times a_X$ preserves the top cycle Γ. Let $\gamma = \mathrm{cl}(\Gamma) \in H^8(A \times X, \mathbf{Q})$, and let p_A, p_X be the projections from $A \times X$. The correspondence action is

$$\Gamma_* u = p_{X*}(\gamma \smile p_A^* u).$$

Since $h_*[\Gamma] = [\Gamma]$, one has $h^*\gamma = \gamma$, and hence

$$(\mathrm{id}_A \times a_X)^* \gamma = (\tilde{\delta}_C^{-1} \times \mathrm{id}_X)^* \gamma.$$

For $u \in H^4(A, \mathbf{Q})$, proper base change along a_X gives

$$a_X^* \Gamma_* u = p_{X*}\big((\mathrm{id}_A \times a_X)^* \gamma \smile p_A^* u\big) = p_{X*}\big((\tilde{\delta}_C^{-1} \times \mathrm{id}_X)^* (\gamma \smile p_A^* \delta^* u)\big) = \Gamma_* \delta^* u.$$

Here $(\tilde{\delta}_C^{-1})^* \delta^* = \mathrm{id}$, and $p_X \circ (\tilde{\delta}_C^{-1} \times \mathrm{id}_X) = p_X$. Therefore $a_X^* \circ \Gamma_* = \Gamma_* \circ \delta^*$, so the matching preserves rather than inverts the character label.

Use pullback characters on both sides. For $c = (c_0, \dots, c_5) \in (\mathbf{Z}/33)^6$ with $\sum_i c_i \equiv 0 \pmod{33}$, write

$$\chi_c([\epsilon_0, \dots, \epsilon_5]) = \prod_{i=0}^{5} \epsilon_i^{c_i}, \qquad H(c) = \{h \in H^4_{\mathrm{prim}}(X_{33}^4, \mathbf{C}) : b^* h = \chi_c(b) h \text{ for all } b \in A\}.$$

On the curve side put $V_n = \{v : \delta^* v = \zeta_{33}^n v\}$. These are pullback labels; under the covariant K-action normalized after (1.0), ζ_{33} acts on V_n by ζ_{33}^{-n}. Write $U = \bigoplus_{n \in (\mathbf{Z}/33)^\times} U_n$, where $U_n = \bigwedge^4 V_n$ is the one-dimensional source character line. Under the K"unneth realization $U_n \subset V_n^{\otimes 4}$, the residual class of $(\delta, 1, 1, 1)$ acts on the first tensor factor only. Since $\delta^*|_{V_n} = \zeta_{33}^n \,\mathrm{id}$, this action on U_n is the scalar ζ_{33}^n. Since $K_\alpha = \ker \chi_\alpha$ and χ_α is surjective, the characters of A trivial on K_α are precisely $r\alpha$, $r \in \mathbf{Z}/33$. The Fermat residue-character decomposition has a one-dimensional primitive line for each character tuple with all coordinates nonzero and sum $0 \pmod{33}$. For $n \in (\mathbf{Z}/33)^\times$, the tuple $n\alpha$ has these properties since every $\alpha_i \neq 0 \pmod{33}$ and $\sum_i \alpha_i = 99$. Its residual eigenvalue is $\chi_{n\alpha}(a) = \zeta_{33}^n$. Thus the K_α-invariant primitive summand of residual character ζ_{33}^n is exactly $H(n\alpha)$. The only nonprimitive summand of $H^4(X_{33}^4, \mathbf{C})$ is the ambient line $\mathbf{C}h^2$, on which A acts trivially. Since n is a unit, $\zeta_{33}^n \neq 1$, so the residual eigenspace in question has no ambient component. Consequently the full K_α-invariant ζ_{33}^n-eigenspace in $H^4(X_{33}^4, \mathbf{C})$ is exactly $H(n\alpha)$. For each $k \in K_\alpha$, the automorphism $\mathrm{id}_A \times k$ preserves W and its top cycle, hence $(\mathrm{id}_A \times k)^* \gamma = \gamma$. Therefore, for $u \in H^4(A, \mathbf{C})$,

$$k^* \Gamma_* u = p_{X*}\big((\mathrm{id}_A \times k)^* \gamma \smile p_A^* u\big) = \Gamma_* u.$$

Thus $\Gamma_* u$ is K_α-invariant. Equivariance gives $\Gamma_* U_n \subseteq H(n\alpha)$. Consequently

$$M_\alpha \otimes \mathbf{C} = \bigoplus_{n \in (\mathbf{Z}/33)^\times} H(n\alpha), \qquad (\Gamma_* U) \otimes_{\mathbf{Q}} \mathbf{C} \subseteq M_\alpha \otimes_{\mathbf{Q}} \mathbf{C}.$$

The correspondence Γ_* is injective on U by (4.4), since $U \subset H^4(A, \mathbf{Q})^G$. Each U_n and each $H(n\alpha)$ is one-dimensional, so the character-wise containments are equalities. Hence $\Gamma_* U \otimes \mathbf{C} = M_\alpha \otimes \mathbf{C}$. Since Γ_*, U, and M_α are rational, this descends to a rational Hodge isomorphism

$$\Gamma_*|_U \,:\, U \xrightarrow{\sim} M_\alpha.$$

The cohomological inverse is explicit. Put

$$P_U := p_G e_{33}^{\boxtimes 4}, \qquad S_\Gamma := P_U \, {}^t\Gamma_*|_{M_\alpha} \,:\, M_\alpha \longrightarrow U.$$

With $\lambda = |G||K_\alpha|$, (4.4) gives

$$S_\Gamma \circ \Gamma_*|_U = \lambda \, \mathrm{id}_U, \qquad S_\Gamma = \lambda \, (\Gamma_*|_U)^{-1}.$$

Here $|G| = 24 \cdot 33^3$ and $|K_\alpha| = 33^4$, so $\lambda = 24 \cdot 33^7$. This is a cohomological inverse induced by algebraic correspondences; no Chow-cycle composition identity is asserted.

The ρ-fixed retraction is

$$R_\Gamma := P_U^\rho \, {}^t\Gamma_*|_{M_\alpha} \,:\, M_\alpha \longrightarrow U^\rho$$

and $P_U^\rho = p_\rho P_U$. Thus $R_\Gamma = p_\rho S_\Gamma$ is surjective. In particular, since P_U^ρ is the identity on U^ρ, (4.4) gives

$$R_\Gamma \circ \Gamma_*|_{U^\rho} = \lambda \, \mathrm{id}_{U^\rho}.$$

Thus $\lambda^{-1}\Gamma_*|_{U^\rho}$ is a right inverse of R_Γ. By (3.1), every class in M_α has an algebraic representative; applying S_Γ shows that all classes in U, and hence in U^ρ, are algebraic. The resulting cycles and correspondences are defined over a common finite extension L after enlarging $K' = \mathbf{Q}(\zeta_{165}, \sqrt[3]{5})$.

5. The curve-to-Prym correspondence

Let $a_D : D \to J(D)$ be the fixed based Abel map and write

$$\nu_D := (a_D^*)^{-1} \,:\, H^1(D, \mathbf{Q}) \xrightarrow{\sim} H^1(J(D), \mathbf{Q}).$$

The quotient correspondences below act first on $H^1(D)$; when comparing their images with i^*, r^*, or endomorphisms of $J(D)$, we use the explicit identification ν_D.

Put

$$V = e_{33} H^1(C, \mathbf{Q}), \qquad H = V^\rho, \qquad H_B := q_* H. \tag{5.1}$$

The finite quotient identities

$$q^* q_* = \sum_{j=0}^{4} \rho^j, \qquad q_* q^* = 5$$

as divisor-correspondence identities. Indeed, the top cycle of $C \times_D C$ is $\sum_{j=0}^{4} \Gamma_{\rho^j}$, with each component of generic multiplicity one, while the trace–pullback composition on D is $5\Delta_D$. Intersections among the graphs lie over finitely many ramification points and contribute no divisor component. Thus the formulas apply to this branched quotient, not only to an étale cover. They also identify the image with the exact-conductor factor. Write $\Pi_B = \frac{1}{5} q_* e_{33} q^*$ for the induced projector on $H^1(D, \mathbf{Q})$. For $x \in H$,

$$\Pi_B q_* x = \frac{1}{5} q_* e_{33} \Big(\sum_{j=0}^{4} \rho^j \Big) x = q_* x,$$

since $e_{33} x = x$ and $\rho x = x$. Conversely, for $y \in H^1(D, \mathbf{Q})$, $q^* y$ is ρ-invariant and e_{33} commutes with ρ, so $\Pi_B y \in q_* H$. Hence $\mathrm{Im}(\Pi_B) = q_* H = H_B$. Moreover, $q_*|_H$ is injective because $q^* q_* x = 5x$ for $x \in H$; thus $q_*|_H : H \xrightarrow{\sim} H_B$. Put

$H_B^J := \nu_D(H_B) \subset H^1(J(D), \mathbf{Q})$. Let $\theta : K \hookrightarrow \mathrm{End}_{\mathbf{Q}}(V)$ be the deck action, in the same covariant graph convention as in (1.0), so $\theta(\zeta_{33}) = \Gamma_\delta = (\delta^{-1})^*$. Let $\gamma(\zeta_{33}) = \zeta_{33}^4$. The unstarred ρ below denotes $\Gamma_\rho = (\rho^{-1})^*$, and hence

$$\rho\theta(\zeta_{33})\rho^{-1} = \theta(\zeta_{33}^4).$$

In the pullback-eigenline notation V_n of Section 4, $\theta(\zeta_{33})$ therefore acts by ζ_{33}^{-n}. For $v \in V_n$, the same conjugacy gives $\theta(\zeta_{33})\rho v = \rho\theta(\zeta_{33}^{25})v = \zeta_{33}^{-25n}\rho v$, hence $\rho(V_n) = V_{25n}$. This is the reverse generator of the same $\langle 4\rangle$-orbit. If $n \in (\mathbf{Z}/33)^\times$ and $\sigma_n : K \hookrightarrow \mathbf{C}$ is given by $\sigma_n(a) = a(\zeta_{33}^{-n})$, then $\sigma_{25n}|_E = \sigma_n|_E$: the five components are only reindexed over the same E-embedding. The synchronous average is unchanged, so the descent factor remains $5^{1-4} = 5^{-3}$. For $a \in E = K^{\langle\gamma\rangle}$,

$$\rho\theta(a)\rho^{-1} = \theta(\gamma(a)) = \theta(a).$$

For $x \in H^1(C)$, define $\tilde\theta(a)x = \theta(a)(e_{33}x)$; this extends $\theta(a)$ by zero on the complementary conductor summands. Since e_{33} is ρ-stable, $\tilde\theta(a)$ also commutes with ρ. The induced operator on $H^1(D)$ is

$$\bar\theta(a) = \frac{1}{5} q_* \tilde\theta(a) q^*.$$

The quotient identities give the intertwining relation

$$\bar\theta(a) q_* = \frac{1}{5} q_* \tilde\theta(a) q^* q_* = \frac{1}{5} q_* \tilde\theta(a) \sum_{j=0}^{4} \rho^j = q_* \tilde\theta(a),$$

where $q_*\rho = q_*$ and $\tilde\theta(a)\rho = \rho\tilde\theta(a)$. Thus $q_*|_H$ is E-linear, and its image $H_B = q_*H$ is E-stable. Each $\bar\theta(a)$ has image in H_B, since q^*y is ρ-invariant and $\tilde\theta(a)$ preserves H; hence $\Pi_B\bar\theta(a) = \bar\theta(a)$. Because $q_*|_H : H \to H_B$ is an isomorphism, the intertwining identity also gives

$$\bar\theta(a)|_{H_B} = (q_*|_H)\,\theta(a)|_H\,(q_*|_H)^{-1}.$$

Thus $a \mapsto \bar\theta(a)|_{H_B}$ is a faithful unital $\mathbf{Q}$-algebra embedding $E \hookrightarrow \mathrm{End}_{\mathbf{Q}}(H_B)$. The other order follows from $q^*q_* = \sum_j \rho^j$, $q_*\rho = q_*$, and $\tilde\theta(a)\rho = \rho\tilde\theta(a)$. Thus

$$\bar\theta(a)\Pi_B = \Pi_B\bar\theta(a) = \bar\theta(a).$$

Transport these operators to $H^1(J(D))$ by setting

$$\Pi_B^J := \nu_D\Pi_B a_D^*, \qquad \bar\theta_J(a) := \nu_D\bar\theta(a)a_D^*.$$

These are the induced actions of the same algebraic curve correspondences on $H^1(J(D))$. Thus the algebraic idempotent defining B_8 commutes with the induced E-action on $H^1(J(D))$; the cohomological representation of $\mathrm{End}^0(J(D))$ is faithful, so this is commutation in $\mathrm{End}^0(J(D))$. The idempotent splits in the isogeny category, and this splitting is E-compatible. After clearing denominators choose i, r, N with $ri = [N]_{B_8}$ and $\pi_B = N^{-1}ir$. Its cohomological pullback is $\pi_B^* = N^{-1}r^*i^* = \Pi_B^J$, so $H_B^J = r^*H^1(B_8, \mathbf{Q})$. Define the target action by

$$\theta_{B_8}(a) = (r^*)^{-1}\,\bar\theta_J(a)\,r^* \quad \text{on } H^1(B_8, \mathbf{Q}).$$

This is exactly the embedding in (1.0b): $\bar\theta_J(a)$ descends the same covariant exact-conductor deck correspondence through the quotient, with no precomposition by $a \mapsto a^{-1}$. The intertwining identity above then gives

$$(r^*)^{-1}\nu_D q_*\theta(a)x = \theta_{B_8}(a)(r^*)^{-1}\nu_D q_* x \qquad (x \in H).$$

Thus $\bar q_*$ intertwines the specified source and target E-actions. Consequently

$$\bar q_* := (r^*)^{-1} \circ \nu_D \circ q_*|_H : H \xrightarrow{\sim}_E H^1(B_8, \mathbf{Q}).$$

We realize this single-factor map by a divisor correspondence. Let $q_J^* : J(D) \to J(C)$ be the Jacobian pullback and put

$$\phi = q_J^* \circ i : B_8 \longrightarrow J(C).$$

After enlarging L if necessary, choose an Abel base point on C and let $\mathcal{L}_C$ be the normalized Poincare bundle on $C \times J(C)$. Choose its sign so that its H^1-action agrees with the Abel–Jacobi convention, and set

$$p_0 = c_1((\mathrm{id}_C \times \phi)^*\mathcal{L}_C), \qquad p = N^{-1}p_0 \in CH^1(C \times B_8)_{\mathbf{Q}}. \tag{5.2}$$

Proposition 5.2 (Poincare normalization and E-linearity). The divisor p_0 in (5.2) acts on $H^1(C)$ as $i^* \circ \nu_D \circ q_*$. Therefore

$$p_*|_H = N^{-1} i^* \nu_D q_*|_H = (r^*)^{-1} \nu_D q_*|_H = \bar{q}_*, \tag{5.2a}$$

and this map is E-linear.

Proof. Let $a_C : C \to J(C)$ be the based Abel map. The pullbacks a_C^* and a_D^* on H^1 are integral isomorphisms. Write $\ell_C : H^1(C) \to H^1(J(C))$ for the correspondence action of $c_1(\mathcal{L}_C)$. The $H^1(C) \otimes H^1(J(C))$ Kunneth component of the normalized Poincare class is the universal evaluation tensor: it is the identity tensor under the principal-polarization identification $J(C) \simeq J(C)^\vee$. Equivalently, if θ_C is the principal polarization class and $g = \dim J(C)$, its action satisfies

$$\int_{J(C)} \ell_C(\alpha) \smile \eta \smile \frac{\theta_C^{g-1}}{(g-1)!} = \varepsilon \int_C \alpha \smile a_C^* \eta \qquad (\alpha \in H^1(C),\ \eta \in H^1(J(C))).$$

The principal-polarization pairing is unimodular and normalized by $\int_{J(C)} \theta_C^g / g! = 1$. Pulling the universal tensor back by a_C gives $a_C^* \ell_C = \varepsilon \, \mathrm{id}_{H^1(C)}$, where $\varepsilon \in \{1, -1\}$ records the Poincare-bundle convention. The resulting map has coefficient one, not a genus or degree multiple. By the sign choice in (5.2), $\varepsilon = 1$, hence $\ell_C = (a_C^*)^{-1}$.

The pullback homomorphism $q_J^* : J(D) \to J(C)$ is induced by pullback of degree-zero divisor classes. Its map on homology is the transfer associated to the transpose of the divisor correspondence of q. Indeed, for a base point $y_0 \in D$, $q_J^* \circ a_D$ sends y to the Abel–Jacobi class of $q^*(y - y_0) \in \mathrm{Div}^0(C)$. The fiber divisor $q^* y$ includes the ramification multiplicities, so this identity also holds at branch values. Since the Abel maps induce integral isomorphisms on H_1, the induced homology map is the transfer $q^! : H_1(D, \mathbf{Q}) \to H_1(C, \mathbf{Q})$. Dualizing under evaluation, equivalently using the curve intersection pairings, gives the cohomological trace:

$$a_D^* \circ (q_J^*)^* \circ (a_C^*)^{-1} = q_*.$$

The normalization is fixed by the adjunction identity

$$\int_D q_* \alpha \smile \beta = \int_C \alpha \smile q^* \beta \qquad (\alpha \in H^1(C),\ \beta \in H^1(D)).$$

The degree $\deg(q)$ appears in the separate composition $q_* q^* = \deg(q)\,\mathrm{id}$, not in this trace map. Since $\phi = q_J^* \circ i$, functoriality of correspondence actions now gives

$$p_{0*} = i^* \circ (q_J^*)^* \circ \ell_C = i^* \circ \nu_D \circ q_*.$$

Finally, $ri = [N]_{B_8}$ implies $i^* r^* = (ri)^* = [N]^* = N$ id on $H^1(B_8)$, giving the displayed formula. The E-linearity follows from the E-linear map $\nu_D q_*|_H$ and the E-compatible splitting $r^* H^1(B_8) = H_B^J$. □

For $w \in H$, write $\nu_D q_* w = r^* \bar{q}_* w$. Since $i^* r^* = N$ on $H^1(B_8)$, one obtains

$$p_* w = N^{-1} i^* \nu_D q_* w = \bar{q}_* w. \tag{5.3}$$

In particular $p_*|_H$ is an E-linear isomorphism.

Galois descent for the semilinear $\mathbf{Q}(\zeta_{33})/E$-module V gives

$$\kappa_{\mathrm{desc}} : U^\rho = \left(\bigwedge^4_{\mathbf{Q}(\zeta_{33})} V \right)^\rho \xrightarrow{\sim} \bigwedge^4_E H.$$

The normalization does not require cyclicity. Let K/E be a finite Galois extension with group G and degree d, acting semilinearly on V, and set $H_G = V^G$. Write $\kappa_G : (\bigwedge_K^r V)^G \xrightarrow{\sim} \bigwedge_E^r H_G$ for canonical Galois descent. On a free G-orbit of character summands, choose v_i in one summand and put $v_i^{(g)} = g v_i$.

$$D_g = \mathrm{Alt}_r\big(v_1^{(g)} \otimes \cdots \otimes v_r^{(g)}\big), \qquad h_i = \sum_{g \in G} v_i^{(g)}, \qquad u = \frac{1}{d} \sum_{g \in G} D_g,$$

where $\mathrm{Alt}_r = (r!)^{-1}\sum_{\pi\in\mathfrak{S}_r}\mathrm{sgn}(\pi)\tau_\pi$. Fix an embedding $\sigma : E \hookrightarrow \mathbf{C}$. After scalar extension, label the factors of $K \otimes_{E,\sigma} \mathbf{C} \simeq \prod_{g\in G}\mathbf{C}$ by G, and let $e_{i,g}$ be the copy of $e_i \in H_G$ in the g-th summand. Then $h_i = \sum_{g\in G} e_{i,g}$ is the fixed vector corresponding to e_i, and the g-component of their K-exterior product corresponds to D_g. Canonical descent therefore sends $\sum_{g\in G} D_g$ to $e_1 \wedge_E \cdots \wedge_E e_r$, so $\kappa_G(u) = d^{-1}h_1 \wedge_E \cdots \wedge_E h_r$. Suppose $p_* : V \to W$ satisfies $p_* g = p_*$ for every $g \in G$, and $p_*|_{H_G} : H_G \to W$ is E-linear. Write $b_i = p_* v_i$, and let T_r apply p_* in each tensor factor followed by cup product. Then $p_* h_i = d b_i$; normalized alternation sends every D_g to the same cup product, giving

$$T_r(u) = b_1 \smile \cdots \smile b_r, \qquad \bigwedge_E^r (p_*|_{H_G})\,\kappa_G(u) = d^{r-1} b_1 \smile \cdots \smile b_r.$$

Consequently, on every free-orbit determinant block,

$$T_r = d^{1-r} \bigwedge_E^r (p_*|_{H_G}) \circ \kappa_G.$$

The identity holds on each free orbit and hence on their sum; it descends to E. In the present case $d = 5$, $r = 4$, and $p_*|_H = \bar{q}_*$, so

$$\bar{q}_*^{(4)} = 5^{-3} \bigwedge_E^4 (\bar{q}_*) \circ \kappa_{\mathrm{desc}} = 5^{-3} \bigwedge_E^4 (p_*|_H) \circ \kappa_{\mathrm{desc}} : U^\rho \xrightarrow{\sim} W_E(B_8). \tag{5.4}$$

The normalized Alt_4 contributes no factorial. Replacing the Poincare bundle by its inverse changes $p_*|_H$ by -1, but leaves (5.4) unchanged because $(-1)^4 = 1$.

6. Small-diagonal determinant transfer

Let $\pi_1^C \in CH^1(C \times C)_{\mathbf{Q}}$ be the degree-one Chow–Kunneth projector and put

$$p_1 = p \circ \pi_1^C, \qquad P = p_1^{\boxtimes 4} \in CH^4(C^4 \times B_8^4)_{\mathbf{Q}}.$$

Let $\Delta_B : B_8 \hookrightarrow B_8^4$ be the small diagonal and define

$$T_B := {}^t\Gamma_{\Delta_B} \circ P. \tag{6.1}$$

The transpose graph has codimension 32 in $B_8^4 \times B_8$. Therefore

$$\mathrm{codim}(T_B) = 4 + 32 - \dim B_8^4 = 4,$$

so

$$T_B \in \mathrm{Corr}^0(C^4, B_8)_{\mathbf{Q}} = CH^4(C^4 \times B_8)_{\mathbf{Q}}. \tag{6.2}$$

In particular,

$$(T_B)_* : CH^2(C^4)_{\mathbf{Q}} \longrightarrow CH^2(B_8)_{\mathbf{Q}}.$$

On the $(1,1,1,1)$-Kunneth block,

$$P_* = p_*^{\otimes 4}, \qquad ({}^t\Gamma_{\Delta_B})_* = \Delta_B^*.$$

Hence

$$(T_B)_* = \Delta_B^* \circ p_*^{\otimes 4} = \mathrm{cup} \circ p_*^{\otimes 4}. \tag{6.3}$$

The scalar in (5.4) can be read off on one orbit. Fix an embedding $\sigma : E \hookrightarrow \mathbf{C}$, a K/E-orbit $\{\tau_0, \dots, \tau_4\}$ above it, and a basis $v_1, \dots, v_4$ of the τ_0-character space. Put $v_i^{(j)} = \rho^j v_i$, let

$$D_j = \mathrm{Alt}_4\big(v_1^{(j)} \otimes \cdots \otimes v_4^{(j)}\big), \qquad h_i = \sum_{j=0}^4 v_i^{(j)} \in H_\sigma(C) := H \otimes_{E,\sigma} \mathbf{C}.$$

Here $\mathrm{Alt}_4 = \frac{1}{4!}\sum_{\pi\in\mathfrak{S}_4}\mathrm{sgn}(\pi)\tau_\pi$; the source projector gives $u_\sigma = \frac{1}{5}\sum_{j=0}^4 D_j$. Under $K \otimes_{E,\sigma} \mathbf{C} \simeq \prod_{j=0}^4 \mathbf{C}$, the determinant $h_1 \wedge_E \cdots \wedge_E h_4$ has components $D_0, \dots, D_4$; equivalently, $\sum_j D_j$ descends to that determinant. Hence

$\kappa_{\mathrm{desc}}(u_\sigma) = \frac{1}{5} h_1 \wedge_E \cdots \wedge_E h_4$. Since $p_*\rho = p_*$, setting $b_i = p_* v_i$ gives $p_* h_i = 5 b_i$. Normalized alternation makes cup product send each D_j to $b_1 \smile \cdots \smile b_4$, hence

$$(T_B)_* u_\sigma = b_1 \smile \cdots \smile b_4, \qquad \bigwedge_E^4 (p_*|_H)\, \kappa_{\mathrm{desc}}(u_\sigma) = 5^3\, b_1 \smile \cdots \smile b_4.$$

Thus the actual small-diagonal map is exactly 5^{-3} times the descended exterior map; no factorial occurs.

After extending scalars to $\mathbf{C}$, the source block U^ρ is the direct sum of four lines indexed by $\sigma : E \hookrightarrow \mathbf{C}$; each line is the synchronous invariant in one K/E-orbit of five determinant character blocks. Write $H_\sigma(C) = H \otimes_{E,\sigma} \mathbf{C}$. Since $p_*|_H$ is an E-linear isomorphism and $p_*\rho = p_*$, it induces

$$p_{*,\sigma} : H_\sigma(C) \xrightarrow{\sim} H_\sigma.$$

On the σ-line, Section 5 identifies T_B, under κ_{desc}, with $5^{-3} \bigwedge_{\mathbf{C}}^4 p_{*,\sigma}$. Since B_8 is a complex torus,

$$H^\bullet(B_8, \mathbf{C}) = \bigwedge^\bullet H^1(B_8, \mathbf{C}), \qquad H^1(B_8, \mathbf{C}) = \bigoplus_{\sigma : E \hookrightarrow \mathbf{C}} H_\sigma.$$

Since $\dim_E H^1(B_8, \mathbf{Q}) = 4$, every H_σ has complex dimension four. Thus $\bigwedge_{\mathbf{C}}^4 H_\sigma$ maps by cup product isomorphically onto a one-dimensional direct summand of $H^4(B_8, \mathbf{C})$; these four summands are pairwise disjoint. Together with the isomorphism $p_{*,\sigma}$, this proves that each source σ-line maps isomorphically to its target determinant line. Their sum is exactly $W_E(B_8)_{\mathbf{C}}$ in the displayed determinant realization, with no other Künneth component; rationality then gives the corresponding statement over $\mathbf{Q}$. The normalized graded alternation defining U^ρ introduces no factorial. Equations (5.3)–(5.4) therefore give

$$(T_B)_*|_{U^\rho} = \bar{q}_*^{(4)} : U^\rho \xrightarrow{\sim} W_E(B_8). \tag{6.4}$$

The finite-quotient adjoint in Section 4 sends algebraic representatives of M_α onto U^ρ. Composing it with T_B therefore sends algebraic cycles on X_{33}^4 to classes spanning $W_E(B_8)$, proving (2.1).

No target determinant projector is required. The source operator P_U^ρ already isolates the exact determinant block, while $(\pi_1^C)^{\boxtimes 4}$ removes every Kunneth component other than $(1,1,1,1)$. Thus every target class produced by the transfer lies in $W_E(B_8)$ itself, rather than merely having a nonzero projection to it.

6.1. A single source-to-target Chow operator

Define

$$\mathcal{S}_B(z_X) = (T_B)_* P_U^\rho\, {}^t\Gamma_* z_X \in CH^2(B_8)_{\mathbf{Q}}, \qquad z_X \in CH^2(X_{33}^4)_{\mathbf{Q}}. \tag{6.6}$$

Equivalently, the underlying degree-zero correspondence is $T_B \circ P_U^\rho \circ {}^t\Gamma$. Every factor is a rational Chow correspondence. Put $R_X = P_U^\rho\, {}^t\Gamma_* : H^4(X_{33}^4, \mathbf{Q}) \to U^\rho$. Then

$$\mathrm{cl}\big(\mathcal{S}_B(z_X)\big) = (T_B)_* R_X(\mathrm{cl}(z_X)). \tag{6.7}$$

Combining Sections 4–6 gives an explicit formula for the cohomological action of (6.6) on the source block. With $\lambda = |G||K_\alpha| = 24 \cdot 33^7$,

$$(T_B)_* R_X|_{M_\alpha} = \lambda\, \bar{q}_*^{(4)} \circ p_\rho \circ (\Gamma_*|_U)^{-1} = \frac{24 \cdot 33^7}{5^3} \bigwedge_E^4 (p_*|_H) \circ \kappa_{\mathrm{desc}} \circ p_\rho \circ (\Gamma_*|_U)^{-1}.$$

Here $(\Gamma_*|_U)^{-1}$ is the cohomological inverse from Section 4, not a Chow-level inverse correspondence. The factor 5^{-3} occurs exactly once, inside $\bar{q}_*^{(4)}$. Since p_ρ projects onto U^ρ and $\bar{q}_*^{(4)} : U^\rho \xrightarrow{\sim} W_E(B_8)$, the image is exactly $W_E(B_8)$. Under the full-block input proved in Section 6.2 (also supplied independently by Aoki), (6.6) therefore sends algebraic cycles representing M_α to cycles spanning $W_E(B_8)$, without an exact Chow-composition identity, Abel–Fourier, refined Gysin, or a target determinant projector. The minimal source-line argument below needs only one nonzero class in the image of R_X.

The marked Kummer data are defined over $K_0 = \mathbf{Q}(\zeta_{165})$. The common-quotient normalization uses

$$c = \zeta_5^4\, 5^{-22}, \qquad d = \zeta_5^3(\sqrt[3]{5})^{-2}, \qquad d^{33} = c,$$

so it is defined over $K' = K_0(\sqrt[3]{5})$. The factor B_8, the morphisms i, r, q_J^*, the Poincare correspondence p, the source cycles, the remaining transfer correspondences, and the finite set of endomorphisms used to span the determinant line are all defined over one finite extension L/K', after enlarging L once and fixing an embedding $L \hookrightarrow \mathbf{C}$. The subscript $\mathbf{Q}$ denotes Chow coefficients, not descent to $\mathbf{Q}$.

6.2. Source image criterion and rank

Define

$$R_X := P_U^{\rho}\, {}^t\Gamma_* : H^4(X_{33}^4, \mathbf{Q}) \longrightarrow U^{\rho}. \tag{6.8}$$

By Section 4, $R_X|_{M_\alpha} : M_\alpha \to U^{\rho}$ is surjective, so R_X has rank 4, and (6.7) holds. Put

$$I_X := \mathrm{Im}\Big(CH^2(X_{33}^4)_{\mathbf{Q}} \xrightarrow{\mathrm{cl}} H^4(X_{33}^4, \mathbf{Q})\Big).$$

The exact criterion for the correspondence (6.6) to span $W_E(B_8)$ is $R_X(I_X) = U^{\rho}$. A sufficient source-block condition is:

$$M_\alpha \subset \mathrm{Im}\Big(CH^2(X_{33}^4)_{\mathbf{Q}} \xrightarrow{\mathrm{cl}} H^4(X_{33}^4, \mathbf{Q})\Big), \tag{6.11}$$

The Aoki source result supplies (6.11) by (3.1), and Section 4 gives $R_X(M_\alpha) = U^{\rho}$. This sufficient condition is not asserted to be necessary. No algebraicity of all Hodge classes of X_{33}^4 is used.

For algebraicity of $W_E(B_8)$, the full-span condition is stronger than necessary. It suffices that the algebraic source classes have a nonzero projection to the single character line $H(-\alpha)$:

$$\mathrm{pr}_{H(-\alpha)}(I_X \otimes_{\mathbf{Q}} \mathbf{C}) \neq 0. \tag{6.12}$$

The character projector is a complex linear combination of deck-automorphism correspondences, so it preserves $I_X \otimes \mathbf{C}$. Indeed, the order-five action permutes U_n by $n \mapsto 25n$, and $\mathcal{O}_1 = \{1, 25, 31, 16, 4\}$ is one orbit. Since the action cycles the five one-dimensional U_n's, its invariant subspace is a line with nonzero projection to every U_n. By the finite-quotient pairing and residual-character matching of Section 4, Γ_* is injective on each U_n and maps it into $H(n\alpha)$. Thus any nonzero vector $u_{\mathcal{O}_1}$ on this invariant line has an image with a nonzero $H(\alpha)$-component. A nonzero class $z \in \mathrm{pr}_{H(-\alpha)}(I_X \otimes \mathbf{C})$ pairs nontrivially with that component and trivially with the other character components: the diagonal group preserves the Poincare pairing, so $H(c)$ is orthogonal to $H(d)$ unless $c + d = 0$; nondegeneracy on primitive H^4 and one-dimensionality give a perfect pairing $H(-\alpha) \times H(\alpha) \to \mathbf{C}$. Since $R_X = P_U^{\rho}\, {}^t\Gamma_*$ is the Poincare adjoint,

$$\langle R_X z, u_{\mathcal{O}_1} \rangle = \langle z, \Gamma_* u_{\mathcal{O}_1} \rangle \neq 0.$$

Hence $R_X(I_X \otimes \mathbf{C}) \neq 0$, and rationality of R_X gives $R_X(I_X) \neq 0$. Choose $z_X \in CH^2(X_{33}^4)_{\mathbf{Q}}$ with $R_X(\mathrm{cl}(z_X)) \neq 0$. The class of the composite cycle in (6.6) is

$$\mathrm{cl}\big(\mathcal{S}_B(z_X)\big) = (T_B)_* R_X(\mathrm{cl}(z_X)) = \bar{q}_*^{(4)}(R_X(\mathrm{cl}(z_X))) = 5^{-3}\left(\bigwedge_E^4 (p_*|_H) \circ \kappa_{\mathrm{desc}}\right)(R_X(\mathrm{cl}(z_X))) \neq 0. \tag{6.13}$$

Indeed, $R_X(\mathrm{cl}(z_X)) \neq 0$, κ_{desc} is an isomorphism, $p_*|_H$ is an E-linear isomorphism, and $5^{-3} \neq 0$. Thus one source character line gives a nonzero rational algebraic class $\omega \in W_E(B_8)$.

The cubic-scroll construction below proves (6.11) directly, without selecting a scroll with one prescribed nonzero residue coefficient. The full block has Hodge type $(2, 2)$ by the unit-age calculation in Section 1. This source-side result does not remove the fixed model or curve-to-Prym hypotheses from the transfer.

The space $W_E(B_8) = \bigwedge_E^4 H^1(B_8, \mathbf{Q})$ has rank one over $E = \mathrm{End}_{\mathbf{Q}}^0(B_8)$. Its E-scalar action can be realized by inserting an endomorphism in just one of the four factors before the small diagonal. For $a \in E$, choose a rational Chow correspondence $\Lambda_a \in CH^8(B_8 \times B_8)_{\mathbf{Q}}$ whose action on $H^1(B_8)$ is the specified E-action by a. Such a correspondence is

obtained from the graph or transpose graph of an integral multiple of the corresponding endomorphism; if the graph convention gives Rosati conjugation, use $\bar{a}$ instead. Put

$$p_a = \Lambda_a \circ p_1, \qquad T_{B,a} = {}^t\Gamma_{\Delta_B} \circ (p_a \boxtimes p_1 \boxtimes p_1 \boxtimes p_1).$$

Thus $T_{B,a} \in CH^4(C^4 \times B_8)_{\mathbf{Q}}$ and acts from $CH^2(C^4)_{\mathbf{Q}}$ to $CH^2(B_8)_{\mathbf{Q}}$. On the determinant line indexed by $\sigma : E \hookrightarrow \mathbf{C}$, the first factor is multiplied by $\sigma(a)$. Since $a \in E = K^{\langle\gamma\rangle}$, this scalar is constant along its five-character orbit. With the orbit notation of Section 5, this reads

$$(T_{B,a})_* u_\sigma = \sigma(a)\, b_1 \smile b_2 \smile b_3 \smile b_4 = \sigma(a)(T_B)_* u_\sigma.$$

Canonical descent therefore gives

$$(T_{B,a})_*|_{U^\rho} = \mu_a \circ (T_B)_*|_{U^\rho}, \qquad \mu_a(\omega) = a\omega.$$

Since $W_E(B_8)$ is an E-line, multiplication $E \to W_E(B_8)$, $a \mapsto a\omega$, is a $\mathbf{Q}$-linear isomorphism for every nonzero ω. Hence the classes $a\omega$, as a ranges over a $\mathbf{Q}$-basis of E, form a $\mathbf{Q}$-basis of $W_E(B_8)$. For example, with $u^2 = -3$ and $v^2 = -11$, $(1, u, v, uv)$ is such a basis. The corresponding cycles are

$$Z_a = (T_{B,a})_* P_U^\rho \, {}^t\Gamma_* z_X \in CH^2(B_8)_{\mathbf{Q}}.$$

For the seed z_X in (6.13), they satisfy $\mathrm{cl}(Z_a) = a\omega$. After enlarging L to define the correspondences and the seed cycle, this finite family is defined over one finite extension. It gives a correspondence-defined basis, but not explicit Chow equations for the seed cycle. The Aoki argument above is an independent proof of the full-block statement (6.11). The cubic-scroll construction below proves the same statement by mapping the full unit orbit; the transfer itself can use only the weaker condition (6.12).

A cubic-scroll realization of the source block Set

$$F_{\mathrm{cub}} = y_0^2 y_1 + y_1^2 y_2 + y_2^2 y_3 + y_3^2 y_4 + y_4^2 y_0 + y_5^3$$

and let $Y_{\mathrm{cub}} \subset \mathbf{P}^5_y$ be its zero locus. Under the permutation $(w_0, w_1, w_2, w_3, w_4, w_5) = (y_5, y_0, y_3, y_2, y_4, y_1)$, this is the order-11 symplectic cubic F_{11}^1 of Billi–Grossi–Marquand [21]. Their family has dimension zero, and their Proposition 4.1 says that its 6270 cubic-scroll classes $[T_i]$ generate $A(Y_{\mathrm{cub}})$.

For a direct check of the geometric input, the partial derivatives are $2y_i y_{i+1} + y_{i-1}^2$ for cyclic indices $i \in \mathbf{Z}/5$ and $3y_5^2$. At a common zero, $y_5 = 0$. If one of $y_0, \dots, y_4$ vanishes, the derivative equations force all five to vanish; otherwise, multiplying $y_{i-1}^2 = -2y_i y_{i+1}$ over all five indices gives $(\prod_i y_i)^2 = -32(\prod_i y_i)^2$, impossible in characteristic zero. Thus Y_{cub} is smooth. It also has the explicit diagonal automorphism

$$(y_0, y_1, y_2, y_3, y_4, y_5) \longmapsto (\zeta_{11} y_0, \zeta_{11}^9 y_1, \zeta_{11}^4 y_2, \zeta_{11}^3 y_3, \zeta_{11}^5 y_4, y_5).$$

Every monomial of F_{cub} has weight zero modulo 11, and the determinant is $\zeta_{11}^{1+9+4+3+5} = 1$; hence this order-11 automorphism acts trivially on $H^{3,1}(Y_{\mathrm{cub}})$ and is symplectic.

On $X^\circ = X_{33}^4 \cap (\mathbf{G}_m)^6$, define

$$\phi(x) = (x^{r_0} : \cdots : x^{r_5}), \quad \begin{aligned} r_0 &= (16, -8, 4, -2, 1, 0), & r_1 &= (1, 16, -8, 4, -2, 0), \\ r_2 &= (-2, 1, 16, -8, 4, 0), & r_3 &= (4, -2, 1, 16, -8, 0), \\ r_4 &= (-8, 4, -2, 1, 16, 0), & r_5 &= (0, 0, 0, 0, 0, 11). \end{aligned}$$

All row sums are 11, so ϕ is projectively well-defined on the torus. With cyclic indices on $r_0, \dots, r_4$,

$$2r_i + r_{(i+1) \bmod 5} = 33e_i \quad (0 \le i < 5), \qquad 3r_5 = 33e_5$$

gives $F_{\mathrm{cub}}(\phi(x)) = F_{33}(x)$. Thus $\phi : X^\circ \to Y_{\mathrm{cub}} \cap (\mathbf{G}_m)^6$ is a morphism. To compute the determinant of its exponent matrix on the projective torus and the logarithmic Jacobian scalar, work in the charts $x_5 = y_5 = 1$ and let R be the 5×5 matrix with rows $r_0, \dots, r_4$ restricted to their first five entries. Let P be the cyclic row shift, so that row i of PR is

row $i+1$ of R. The relations above give $(2I+P)R = 33I$, while $\det(2I+P) = \prod_{\zeta^5=1}(2+\zeta) = 33$; hence $\det R = 33^4$. In these charts $\Omega_x = -dx_0 \wedge \cdots \wedge dx_4$ and similarly for Ω_y. The five column sums of R are all 11, so the logarithmic Jacobian, $d\log y_i = \sum_j R_{ij}\, d\log x_j$, gives the exact identity

$$\phi^*\Omega_y = \det(R) \prod_{a=0}^{4} x_a^{11-1}\, \Omega_x = 33^4 \prod_{a=0}^{4} x_a^{10}\, \Omega_x.$$

This determinant is the pullback Jacobian factor; the correspondence does not contribute a second degree factor.

Let $\phi_{\mathbb{T}} : (\mathbf{G}_m)^5_x \to (\mathbf{G}_m)^5_y$ be the induced homomorphism of projective tori in the charts $x_5 = y_5 = 1$. The same matrix identifies its finite kernel. Its Smith normal form is $\mathrm{diag}(1, 33, 33, 33, 33)$: modulo 3, all rows of R are $(1,1,1,1,1)$, while modulo 11 they are successive multiples $1, 9, 9^2, 9^3, 9^4$ of the first row; together with $\det R = 3^4 11^4$, these ranks determine the invariant factors. In particular, $\ker(\phi_{\mathbb{T}})$ has order 33^4. The identity $(2I+P)R = 33I$ also shows that every point of this kernel has all five coordinates in μ_{33}, so it is a subgroup of the projective diagonal group $A = (\mu_{33})^6/\mu_{33,\mathrm{diag}}$, using the chart $x_5 = 1$.

Modulo 33, the rows of R are respectively $1, 31, 4, 25, 16$ times $c = (16, 25, 4, 31, 1)$; these are units, so the pulled-back target character lattice is generated by c. Its projective lift $\tilde{c} = (16, 25, 4, 31, 1, 22)$ satisfies $\tilde{c} = 13\beta$, with $13 \in (\mathbf{Z}/33)^\times$. Hence the annihilator of $\ker(\phi_{\mathbb{T}})$ in A's character group is generated by χ_β, and

$$\ker(\phi_{\mathbb{T}}) = K_\beta := \ker \chi_\beta.$$

Writing $Y^\circ_{\mathrm{cub}} = Y_{\mathrm{cub}} \cap (\mathbf{G}_m)^6$, the identity $F_{\mathrm{cub}} \circ \phi = F_{33}$ gives $X^\circ = \phi_{\mathbb{T}}^{-1}(Y^\circ_{\mathrm{cub}})$. Thus $X^\circ \to Y^\circ_{\mathrm{cub}}$ is a finite étale K_β-torsor of degree 33^4, and $Y^\circ_{\mathrm{cub}} \simeq X^\circ/K_\beta$.

Under Griffiths' residue isomorphism [22], the twenty classes

$$\omega_{ijk} = \mathrm{Res}_{Y_{\mathrm{cub}}}\left(\frac{y_i y_j y_k\, \Omega_y}{F_{\mathrm{cub}}^3}\right), \qquad 0 \le i < j < k \le 5,$$

lie in $H^{2,2}_{\mathrm{prim}}(Y_{\mathrm{cub}}, \mathbf{C})$. In the Jacobian ring the relations are $y_i^2 = -2y_{i+1}y_{i+2}$ for $i \in \mathbf{Z}/5$ and $y_5^2 = 0$. They reduce every cubic monomial to a linear combination of the twenty squarefree monomials: for $j = i+2$, $y_i^2 y_j = 4y_{i+1}y_{i+3}y_{i+4}$; for $j = i+3, i+4$ one reduction suffices; $y_i^3 = -2y_i y_{i+1} y_{i+2}$; and $y_i^2 y_{i+1} = 0$, since cycling the relations gives $y_i^2 y_{i+1} = -32 y_i^2 y_{i+1}$. Terms containing y_5^2 vanish and $y_5 y_i^2$ reduces to a squarefree monomial. The Jacobian ring has Hilbert series $(1+t)^6$, so its degree-three piece has dimension 20; hence these twenty residues are a basis.

For each triple define

$$\beta_{ijk} = 11(1,1,1,1,1,1) + r_i + r_j + r_k \pmod{33}; \qquad \beta_{045} = \beta.$$

The following table gives the unique unit $u_{ijk} \in (\mathbf{Z}/33)^\times$ with $\beta_{ijk} = u_{ijk}\beta$:

ijk	012	013	014	015	023
u_{ijk}	17	26	8	31	5
ijk	024	025	034	035	045
u_{ijk}	20	10	29	19	1
ijk	123	124	125	134	135
u_{ijk}	32	14	4	23	13
ijk	145	234	235	245	345
u_{ijk}	28	2	25	7	16

Let Γ_ϕ be the closure of the graph and, for $\eta \in H^4(Y_{\mathrm{cub}}, \mathbf{Q})$, write

$$C_\phi(\eta) = (p_X)_*\big([\Gamma_\phi] \cdot p_Y^*\eta\big) \in H^4(X_{33}^4, \mathbf{Q})$$

for its cohomological pullback correspondence, extended linearly to complex coefficients. Over X°, the graph closure restricts to the graph of ϕ, so this correspondence acts there by the ordinary pullback ϕ^*. These twenty units are exactly $(\mathbf{Z}/33)^\times$, so the characters are distinct and exhaust the orbit $[\beta]$. The row identities, determinant and Smith

calculation, residue bounds, and complete unit table are checked by the companion exact computation. That check covers only the finite arithmetic; the geometric residue and boundary-support arguments are proved in the text. The coordinate permutation $(\beta_1,\beta_3,\beta_2,\beta_0,\beta_5,\beta_4)=\alpha$ identifies β with α; hence the unit-age identity of Section 1 gives $\sum_a\langle(\beta_{ijk})_a\rangle_{33}=99$ for every triple. The raw homogeneous exponent vector in the pullback is

$$e_{ijk}=10\mathbf{1}+r_i+r_j+r_k,\qquad \mathbf{1}=(1,1,1,1,1,1),\qquad e_{ijk}+\mathbf{1}\equiv\beta_{ijk}\pmod{33}.$$

For each coordinate $a<5$, the entries of $r_0,\dots,r_4$ form the multiset $\{-8,-2,1,4,16\}$. If the triple omits 5, this gives $1\le(e_{ijk})_a\le 31$; if it contains 5, it gives $0\le(e_{ijk})_a\le 30$. The sixth coordinate is respectively 10 or 21. Therefore every coordinate of e_{ijk} lies in $[0,31]$, so $1\le(e_{ijk})_a+1\le 32$ for every a. This is already the least positive residue of $(\beta_{ijk})_a$, with no reduction wraparound. The residue monomial has degree $\sum_a(e_{ijk})_a=60+3\cdot 11=93$, and the logarithmic pullback formula is

$$C_\phi(\omega_{ijk})|_{X^\circ}=33^4\operatorname{Res}_{X^\circ}\left(\frac{\prod_{a=0}^5 x_a^{\langle(\beta_{ijk})_a\rangle_{33}-1}\,\Omega_x}{F_{33}^3}\right).$$

The residue monomial has degree $93=3\cdot 33-6$, with each exponent between 0 and 31, so it is nonzero in the Fermat Jacobian ring. Its global residue class $r_{ijk}\in H(\beta_{ijk})$ is therefore nonzero.

Put $D_i=X_{33}^4\cap\{x_i=0\}$ and $D_I=\bigcap_{i\in I}D_i$ for nonempty $I\subseteq\{0,\dots,5\}$. If $1\le|I|\le 4$, then D_I is the smooth Fermat hypersurface in $\mathbf{P}^{5-|I|}$ and has codimension $|I|$ in X_{33}^4; if $|I|\ge 5$, it is empty. Indeed, at every point of a nonempty D_I, some $x_j\neq 0$ with $j\notin I$, so the Fermat gradient $\partial F_{33}/\partial x_j=33x_j^{32}$ has a nonzero coordinate outside the coordinate normals dx_i, $i\in I$. Thus the D_i form a simple-normal-crossing divisor. Let $A=(\mu_{33})^6/\mu_{33,\mathrm{diag}}$ be the effective diagonal automorphism group of X_{33}^4, and let $A_i\subset A$ be the image of the i-th coordinate subgroup μ_{33}; this image is again μ_{33}, since the coordinate subgroup meets the diagonal subgroup trivially. If $i\in I$, then A_i fixes D_I pointwise. On the normal bundle $N_{D_I/X_{33}^4}\simeq\bigoplus_{j\in I}\mathcal{O}_{D_I}(1)$, it acts by a complex-linear scalar on the i-th summand and trivially on the others; therefore it preserves the real orientation and the complex Thom class. The Thom isomorphism gives trivial A_i-action on $H^*_{D_I}(X_{33}^4,\mathbf{C})$.

The A-equivariant Mayer–Vietoris spectral sequence for supports in $D=\bigcup_i D_i$ is built from the groups $H^*_{D_I}(X_{33}^4,\mathbf{C})$, $I\neq\varnothing$. Each such term is trivial under at least one coordinate subgroup A_i, so it contains no character nontrivial in all six coordinates. Every $H(u\beta)$ is nontrivial on each A_i, since its i-th character exponent is $u\beta_i\not\equiv 0\pmod{33}$. Thus no $H(u\beta)$ occurs on this first page. Since $\mathbf{C}[A]$ is semisimple and the differentials are A-equivariant, the same vanishing holds in $H^4_D(X_{33}^4,\mathbf{C})$. The localization sequence therefore makes restriction to X° injective on M_β. Set

$$M_\beta=\bigoplus_{u\in(\mathbf{Z}/33)^\times}H(u\beta).$$

If $e_{[\beta]}$ is the rational projector onto this orbit, the open pullback formula and this injectivity show that each ω_{ijk} maps to its nonzero global residue class r_{ijk}. The twenty images lie in distinct character lines. Since $\dim M_\beta=20$,

$$(e_{[\beta]}\circ C_\phi)_{\mathbf{C}}:H^{2,2}_{\mathrm{prim}}(Y_{\mathrm{cub}},\mathbf{C})\xrightarrow{\sim}M_\beta\otimes_{\mathbf{Q}}\mathbf{C}.$$

By [21, Theorem 3.5 and Table 1], the algebraic lattice $A(Y_{\mathrm{cub}})$ has rank 21, and by [21, Proposition 4.1] it is generated by the 6270 cubic-scroll classes. Write $A_{\mathrm{prim}}(Y_{\mathrm{cub}})_{\mathbf{Q}}$ for the primitive part of this lattice with rational coefficients. It has rank 20, equal to the dimension of $H^{2,2}_{\mathrm{prim}}(Y_{\mathrm{cub}},\mathbf{C})$; hence $A_{\mathrm{prim}}(Y_{\mathrm{cub}})_{\mathbf{Q}}\otimes\mathbf{C}$ is this entire primitive Hodge space. The graph correspondence and the rational orbit projector are algebraic and defined over a finite extension. The displayed isomorphism therefore carries the primitive algebraic lattice onto M_β, proving $M_\beta\subset I_X$. A rational coordinate permutation carries M_β to M_α, giving (6.11), and hence also the minimal condition (6.12). This argument follows the cubic-scroll strategy of Favero–Kelly [23, §4.3], using their monomial map [23, (4.4)] with the cubic equation corrected so that Y_{cub} is smooth (see Section 9).

A finite scroll-indexed family on B_8 Let $h=c_1(\mathcal{O}_{Y_{\mathrm{cub}}}(1))$, and choose the 6270 cubic-scroll classes $[T_i]$ of [21, Proposition 4.1]. Since $\int_{Y_{\mathrm{cub}}}h^4=3$ and $\int_{Y_{\mathrm{cub}}}[T_i]\smile h^2=\deg(T_i)=3$, its primitive projection is $t_i=[T_i]-h^2$. Let $\Gamma_\phi\subset X_{33}^4\times Y_{\mathrm{cub}}$ be the graph closure from the preceding construction, with projections $\mathrm{pr}_X,\mathrm{pr}_Y$, and define its Chow action by

$$\Phi_{\mathrm{cub},*}(t)=(\mathrm{pr}_X)_*\big([\Gamma_\phi]\cdot\mathrm{pr}_Y^*t\big).$$

Let

$$\tau(x_0 : x_1 : x_2 : x_3 : x_4 : x_5) = (x_1 : x_3 : x_2 : x_0 : x_5 : x_4).$$

Since the graph acts covariantly, $\tau_* = (\tau^{-1})^*$; on pullback-character spaces this gives

$$\tau_* H(c_0, c_1, c_2, c_3, c_4, c_5) = H(c_1, c_3, c_2, c_0, c_5, c_4).$$

The character table has $(\beta_1, \beta_3, \beta_2, \beta_0, \beta_5, \beta_4) = \alpha$, so $\tau_* M_\beta = M_\alpha$. Let Γ_τ be its graph, acting by τ_*. The source-to-target correspondence realizing $\Phi_{\mathrm{cub},*}$ is the transposed graph ${}^t\Gamma_\phi \in \mathrm{Corr}^0(Y_{\mathrm{cub}}, X_{33}^4)_{\mathbf{Q}}$; this agrees with the displayed action formula. Set

$$\Sigma_B := T_B \circ P_U^\rho \circ {}^t\Gamma \in \mathrm{Corr}^0(X_{33}^4, B_8)_{\mathbf{Q}}, \qquad \Theta_B := \Sigma_B \circ \Gamma_\tau \circ e_{[\beta]} \circ {}^t\Gamma_\phi \in \mathrm{Corr}^0(Y_{\mathrm{cub}}, B_8)_{\mathbf{Q}}. \tag{6.14}$$

All factors have degree zero, so $\Theta_B \in CH^4(Y_{\mathrm{cub}} \times B_8)_{\mathbf{Q}}$ and acts from $CH^2(Y_{\mathrm{cub}})_{\mathbf{Q}}$ to $CH^2(B_8)_{\mathbf{Q}}$. Although the fiber product defining Γ can be singular, its top-dimensional cycle defines a Chow class on the smooth product $C^4 \times X_{33}^4$; correspondence composition uses that class and does not require the fiber product itself to be smooth. If p_C, p_X are the ambient projections, its action is

$$\Gamma_* z = p_{X*}(p_C^* z \cdot \Gamma), \qquad \mathrm{cl}(\Gamma_* z) = p_{X*}\big(p_C^* \,\mathrm{cl}(z) \smile \mathrm{cl}(\Gamma)\big).$$

Here p_C^* is flat pullback to the smooth product and the intersection is taken there; no pullback to the singular fiber product is used. Put

$$z_i = \tau_*\big((e_{[\beta]})_* \Phi_{\mathrm{cub},*}(t_i)\big) \in CH^2(X_{33}^4)_{\mathbf{Q}}, \qquad Z_i^B = \mathcal{S}_B(z_i) = (\Theta_B)_*(t_i) \in CH^2(B_8)_{\mathbf{Q}} \quad (1 \le i \le 6270).$$

Then, under H1–H4 and the curve-to-Prym correspondence of Sections 2–6,

$$\mathrm{span}_{\mathbf{Q}}\{\mathrm{cl}(Z_i^B) : 1 \le i \le 6270\} = W_E(B_8).$$

Indeed, Proposition 4.1 says that the $[T_i]$ generate $A(Y_{\mathrm{cub}})$, while Theorem 3.5 gives its rank 21. Their primitive projections therefore span the 20-dimensional primitive $(2,2)$-space. The isomorphism $e_{[\beta]} C_\phi$ sends that space onto M_β; τ sends it onto M_α. The adjoint transfer maps M_α onto U^ρ, and the small-diagonal map sends U^ρ isomorphically to $W_E(B_8)$. All operations are algebraic correspondences, so each Z_i^B is a rational algebraic cycle. Since the family is finite, its members and the fixed correspondences are defined over a common finite extension of K'. Since this family spans the four-dimensional target, four linearly independent members can be selected. Proposition 6.1 establishes full character support on the source for each scroll, but not nonvanishing of every unprojected target image. From any selected nonzero member, the endomorphism construction above gives four correspondence-defined cycles spanning the target determinant space; it supplies neither scroll equations nor coordinate Chow equations for the transferred cycles.

The finite-field-of-definition assertion does not require the scrolls initially chosen over $\mathbf{C}$ to have algebraic coordinates. Each scroll determines a complex point of the Hilbert scheme of Y_{cub} over $\overline{\mathbf{Q}}$, in the open locus parameterizing smooth cubic scrolls. The irreducible component over $\overline{\mathbf{Q}}$ containing that point after base change to $\mathbf{C}$ has dense $\overline{\mathbf{Q}}$-points, and the smooth-scroll locus is a nonempty open in it. Choose a $\overline{\mathbf{Q}}$-point in that open. The universal flat family preserves the cohomology class, so the resulting cycle has the same class as the original scroll. The finite list of classes can therefore be represented over one number field; adjoining the fields of definition of the fixed correspondences gives the stated common finite extension.

Proposition 6.1 (full character support of each cubic scroll). Let T_i be any of the 6270 cubic scrolls on Y_{cub}, and put $t_i = [T_i] - h^2$. Then $t_i \neq 0$, and the rational class

$$m_i = e_{[\beta]} C_\phi(t_i) \in M_\beta$$

has nonzero projection to every character line $H(u\beta)$, $u \in (\mathbf{Z}/33)^\times$. Consequently $\mathrm{pr}_{H(-\alpha)}\,\mathrm{cl}(z_i) \neq 0$, so the projected class witnesses the source-line condition (6.12). Under H1–H4, the family $\{\mathrm{cl}(Z_i^B)\}$ spans $W_E(B_8)$, as proved above; this gives at least four linearly independent nonzero members. The proposition does not assert that every unprojected target member is nonzero.

Proof. Identify $T_i \simeq \mathbb{F}_1$, and let C_0, f be its negative section and ruling fiber, so $C_0^2 = -1$, $C_0 \cdot f = 1$, $f^2 = 0$, and $h|_{T_i} = C_0 + 2f$. Then $c_1(T_{T_i}) = 2C_0 + 3f$ and $\int_{T_i} c_2(T_{T_i}) = 4$. The cubic hypersurface has $c(T_{Y_{\mathrm{cub}}}) = (1+h)^6/(1+3h)$, hence $c_1(T_{Y_{\mathrm{cub}}}) = 3h$ and $c_2(T_{Y_{\mathrm{cub}}}) = 6h^2$. The normal sequence gives $c_1(N_{T_i/Y_{\mathrm{cub}}}) = C_0 + 3f$ and

$$\int_{T_i} c_1(T_{T_i})c_1(N_{T_i/Y_{\mathrm{cub}}}) = 7, \qquad \int_{T_i} c_2(T_{Y_{\mathrm{cub}}}|_{T_i}) = 18.$$

Thus $\int_{T_i} c_2(N_{T_i/Y_{\mathrm{cub}}}) = 18 - 4 - 7 = 7$, so $[T_i]^2 = 7$. Since $[T_i] \cdot h^2 = \int h^4 = 3$, the primitive projection $t_i = [T_i] - h^2$ satisfies $t_i^2 = 7 - 2 \cdot 3 + 3 = 4$, in particular $t_i \neq 0$.

By the source isomorphism of Section 6.2, $m_i \neq 0$. It is rational because both $e_{[\beta]}$ and C_ϕ are rational correspondences. The twenty character lines $H(u\beta)$ are distinct and form one Galois orbit, as recorded in the table above. If π_β denotes the character projector, then $\sigma_u(\pi_\beta) = \pi_{u\beta}$ for $\sigma_u \in \mathrm{Gal}(\mathbf{Q}(\zeta_{33})/\mathbf{Q})$; rationality gives $\pi_{u\beta} m_i = \sigma_u(\pi_\beta m_i)$. If one projection vanished, all would vanish, contradicting $m_i \neq 0$ because these lines exhaust $M_\beta \otimes_{\mathbf{Q}} \mathbf{C}$. Thus every projection is nonzero. The coordinate permutation carries this property to every line of M_α, in particular $H(-\alpha)$. Thus the character-projected class of each z_i is a nonzero algebraic witness for (6.12). This does not imply $R_X(\mathrm{cl}(z_i)) \neq 0$: the projector P_U^ρ in R_X averages the five orbit components, which may cancel. Proposition 6.2 shows that the order-five-averaged scroll family spans, although it does not identify a surviving index. From any nonzero target member, the one-factor E-correspondences $T_{B,a}$ of Section 6.2, for a in a $\mathbf{Q}$-basis of E, give a $\mathbf{Q}$-basis of $W_E(B_8)$. □

Proposition 6.2 (order-five averaged scrolls span). Define

$$P_\sigma = \frac{1}{5}\sum_{j=0}^{4} (\sigma^j)_* \quad \text{on } CH^2(Y_{\mathrm{cub}})_{\mathbf{Q}},$$

where $\sigma(y_j) = y_{j-1}$ for $j \in \mathbf{Z}/5$ and $\sigma(y_5) = y_5$. In the Billi–Grossi–Marquand coordinates w, their order-five generator from [21, Proposition 4.3] acts by $\tau(w_0, \dots, w_5) = (w_0, w_2, w_5, w_1, w_3, w_4)$, whereas this σ acts by $(w_0, w_4, w_3, w_5, w_2, w_1) = \tau^3(w)$. Thus $P_\sigma = P_\tau$, since 3 is invertible modulo 5. Use in (6.14) the common-model Kummer normalization whose root λ is chosen as in the proof below. Under H1–H4 and the curve-to-Prym correspondence of Sections 2–6,

$$\mathrm{span}_{\mathbf{Q}}\{\mathrm{cl}(\Theta_B(P_\sigma t_i)) : 1 \le i \le 6270\} = W_E(B_8).$$

Thus at least four averaged members are linearly independent. This is an existence statement; it does not select a canonical scroll index or supply scroll equations.

Proof. The monomial rows satisfy $r_{i-1} = 16r_i \pmod{33}$ for cyclic indices $i \in \mathbf{Z}/5$, and $16r_5 = r_5$. Consequently $\beta_{i-1,j-1,k-1} = 16\beta_{ijk}$, with index 5 fixed, and $\sigma^*\omega_{ijk} = \omega_{i-1,j-1,k-1}$. The cubic action therefore permutes the twenty Fermat character lines by multiplication by 16. On the source, Section 5 gives $\rho_*(U_n) = U_{25n}$, hence the pullback $\rho^{2*} = \rho_*^3$ sends U_n to U_{16n}. Thus the pullback actions match; replacing either generator by its inverse leaves the order-five average unchanged. The common invariant subspace has dimension four.

They also agree geometrically on the common quotient. Scaling the four divisor points by $z \mapsto \zeta_5 z$ gives $Z_i \mapsto Z_{i-1}$ for $0 \le i < 5$, while Z_5 is fixed. In the ordered α-coordinates of Section 2, ρ^* has index map $(1, 3, 5, 2, 4, 0)$, and ρ^{2*} has $(3, 2, 0, 5, 4, 1)$. The displayed β-to-α coordinate permutation transports the cubic five-cycle to this same square. Both lifts to the common degree-33 Kummer cover act on the residual deck group by $\delta \mapsto \delta^{16}$. If the lifts differ by δ^a, the order-five condition gives

$$\delta^{(1+16+16^2+16^3+16^4)a} = \delta^{11a} = 1,$$

so $3 \mid a$. Conjugating a lift by δ^b changes a by $(1-16)b = -15b$, which ranges over all multiples of 3 modulo 33. The common-model normalization $w_C \mapsto \lambda w_F$ may be changed by any deck transformation, so choose it to make the lifts agree. On the dense torus, the cyclic row shifts also give

$$\phi(x_1 : x_2 : x_3 : x_4 : x_0 : x_5) = \sigma^{-1}\phi(x_0 : x_1 : x_2 : x_3 : x_4 : x_5).$$

Thus the graph closure and the full correspondence chain are order-five equivariant. The invariant primitive cubic subspace maps isomorphically to U^ρ, which has dimension four.

The t_i span the primitive algebraic Hodge space by [21, Theorem 3.5, Table 1, and Proposition 4.1]. Applying P_σ to this spanning family spans its invariant subspace. Equivariance and the transfer isomorphism identify that subspace with $W_E(B_8)$, proving the claim. □

7. A balanced special-unitary theorem for all self-powers

The preceding transfer supplies the determinant classes for B_8. The following structural statement isolates the representation-theoretic implication and does not assert determinant algebraicity for an arbitrary abelian variety. No novelty is claimed for it. Weil subspaces attached to field actions and their relation to divisor-generated Hodge rings form a standard framework; the statement below is a special case of the structure theorem of Moonen–Zarhin [3,4] combined with standard invariant theory for the Hodge group, and closely related statements — in particular the implication "the Hodge conjecture for a general abelian variety of Weil type implies it for all of its self-powers" — appear in Milne [9] and in Varesco [6, Proposition 0.4 and Section 3]. We record it here as a self-contained account within that known framework, with a complete proof, only because we need its precise integer-polarization form, which extracts all mixed determinant classes on every power from the rational determinant space of the single factor. The contribution specific to this paper is not the structural statement but the algebraicity of the full rational determinant space for this fixed B_8, established in Sections 3–6.

The October 6, 2026 OpenAI preprint *Weil classes and Hodge classes on abelian powers* claims the Hodge conjecture on all powers of imaginary-quadratic abelian sixfolds with a compatible hyperbolic Hermitian form of signature $(3,3)$, and on all powers of imaginary-quadratic abelian varieties of dimension at most five [24]. Its stated families do not include the dimension-eight variety B_8; its proof also invokes the separate CM-Hodge preprint, which we do not use. We record this as current related public work, not as an independently verified input.

Structural theorem

Let $A/\mathbf{C}$ be a simple abelian variety with $\mathrm{End}^0(A) = E$ a CM field and

$$V = H^1(A, \mathbf{Q}), \qquad \dim_E V = 2m,$$

and assume:

1. the Rosati involution of a polarization restricts to complex conjugation on E;
2. every E-embedding has Hodge signature (m, m);
3. $Hg(A) = \mathrm{Res}_{E^+/\mathbf{Q}} SU_E(V, h)$;
4. the rational determinant space

$$W_E(A) := \bigwedge_E^{2m} V$$

is spanned by classes in $CH^m(A)_{\mathbf{Q}}$.

For $a = (a_1, \dots, a_n) \in \mathbf{Z}^n$, let

$$f_a : A^n \longrightarrow A, \qquad (x_1, \dots, x_n) \longmapsto \sum_j [a_j] x_j,$$

and write

$$\mathcal{P}_n(W_E(A)) = \mathrm{Span}_{\mathbf{Q}}\{f_a^* w : a \in \mathbf{Z}^n,\ w \in W_E(A)\}. \tag{7.1}$$

Then, for every $n \geq 1$, the Hodge ring of A^n is

$$\bigoplus_p H^{2p}(A^n, \mathbf{Q}) \cap H^{p,p}(A^n) = \mathbf{Q}\left\langle \mathrm{NS}(A^n)_{\mathbf{Q}},\ \mathcal{P}_n(W_E(A)) \right\rangle, \tag{7.2}$$

and every class on the right is algebraic. In particular, the classical rational Hodge conjecture holds for every A^n.

Proof The rational Hodge classes on an abelian variety are precisely the invariants of its Hodge group. Since the Hodge structure on $H^1(A^n, \mathbf{Q}) = V \otimes \mathbf{Q}^n$ is a direct sum of copies of V, its Hodge group is the diagonal copy of $Hg(A)$. Consequently the left side of (7.2), after complexification, is

$$\left(\bigwedge^{\mathrm{even}} (V_{\mathbf{C}} \otimes \mathbf{C}^n)\right)^{Hg(A)_{\mathbf{C}}}. \tag{7.2a}$$

Over $\mathbf{C}$, the Hodge group is a product of special linear groups

$$Hg(A)_{\mathbf{C}} \simeq \prod_v SL(W_v), \qquad \dim W_v = 2m,$$

and

$$V_{\mathbf{C}} \simeq \bigoplus_{v} (W_v \oplus W_v^\vee).$$

For $L = \mathbf{C}^n$, the mixed-tensor first fundamental theorem for $SL(W_v)$ [25] gives the tensor invariants. The tensor algebra maps equivariantly and surjectively to the exterior algebra. Since $SL(W_v)$ is reductive in characteristic zero, its invariants functor is exact; hence the induced map on invariants is still surjective. It follows that

$$\left(\dot{\bigwedge} ((W_v \oplus W_v^\vee) \otimes L) \right)^{SL(W_v)}$$

is generated by evaluation contractions and the determinant and dual-determinant tensors. The determinant legs carry repeated copy labels: the Cauchy decomposition

$$\bigwedge^{2m} (W_v \otimes L) = \bigoplus_{\lambda \vdash 2m} S_\lambda W_v \otimes S_{\lambda'} L$$

shows that the only invariant $S_{(1^{2m})} W_v = \det W_v$ contributes $S_{(2m)} L = \mathrm{Sym}^{2m} L$. Thus no distinct-label assumption is hidden in (7.2).

The evaluation contractions are divisor classes. Indeed

$$\mathrm{NS}(A^n)_{\mathbf{Q}} \simeq \mathrm{End}^0(A^n)^{\dagger=1} = M_n(E)^{\dagger=1},$$

and its complexification gives all copy-to-copy contraction spaces: explicitly,

$$M_n(E)^{\dagger=1} \otimes_{\mathbf{Q}} \mathbf{C} \simeq \bigoplus_{\tau : E^+ \hookrightarrow \mathbf{C}} \mathrm{End}_{\mathbf{C}}(L),$$

and the matrix units in each summand give the contractions between every ordered pair of copies of W_τ and $W_\tau^\vee$. For $w \in W_E(A)$, the class $f_a^* w$ is a homogeneous polynomial of degree $2m$ in $a_1, \dots, a_n$; its coefficients are exactly the mixed determinant classes. The explicit finite-difference identity (7.4) below extracts every coefficient from finitely many classes $f_a^* w$, proving that all determinant generators are algebraic. Since the construction starts with the whole rational space $W_E(A)$, rather than an individual complex embedding line, it respects the rational structure (and, in the arithmetic case, Galois descent). This proves (7.2).

More explicitly, put $d = 2m$ and let Φ_w be the homogeneous, cohomology-valued polynomial characterized by

$$\Phi_w(a) = f_a^* w = \sum_{|\nu|=d} c_\nu(w) a^\nu \qquad (a \in \mathbf{Z}^n).$$

For every $\nu \in \mathbf{Z}^n_{\geq 0}$ with $|\nu| = d$, its coefficient is

$$c_\nu(w) = \frac{1}{\nu!} \sum_{0 \leq b \leq \nu} (-1)^{d-|b|} \binom{\nu}{b} f_b^* w, \qquad \nu! = \prod_j \nu_j!, \quad \binom{\nu}{b} = \prod_j \binom{\nu_j}{b_j}. \tag{7.4}$$

Indeed, if $|\mu| = |\nu| = d$ and $\mu \neq \nu$, some j has $\mu_j < \nu_j$, so $\Delta_j^{\nu_j}$ kills the factor $t_j^{\mu_j}$; for $\mu = \nu$, the product of differences at 0 equals $\nu!$. Thus (7.4) is a finite, explicit cycle-class formula for each mixed determinant generator. If $\mathrm{cl}(z_w) = w$, the same combination of $f_b^* z_w$ is a codimension-m cycle representing $c_\nu(w)$. For B_8, $d = 4$: each coefficient uses at most 16 pullbacks (since $\prod_j (\nu_j + 1) \leq 16$) and adds denominator at most $4! = 24$, beyond the denominator of the chosen representative z_w.

Application to B_8

For B_8, $m = 2$, (1.1)–(1.3) verify the structural hypotheses and (2.1) supplies the determinant classes after the fixed embedding $L \hookrightarrow \mathbf{C}$. Consequently (7.2) is the assertion (2.2) in the Main Theorem. For $n = 1$, the invariant Hilbert polynomial is

$$(1 + x + 3x^2 + x^3 + x^4)^2 = 1 + 2x + 7x^2 + 8x^3 + 13x^4 + 8x^5 + 7x^6 + 2x^7 + x^8. \tag{7.3}$$

The exact codimension-two obstruction

Let $D^2(B_8) \subset H^4(B_8, \mathbf{Q})$ be the image of $\mathrm{Sym}^2 \mathrm{NS}(B_8)_{\mathbf{Q}}$ under cup product. The representation in (7.2a) gives

$$Hg(B_8)_{\mathbf{C}} \simeq SL(W_1) \times SL(W_2), \qquad H^1(B_8, \mathbf{C}) = (W_1 \oplus W_1^\vee) \oplus (W_2 \oplus W_2^\vee), \quad \dim W_1 = \dim W_2 = 4.$$

Put $U_j = W_j \oplus W_j^\vee$. For each j, the invariant spaces in degrees two and four are

$$(\bigwedge^2 U_j)^{SL(W_j)} = \mathbf{C}\omega_j, \qquad (\bigwedge^4 U_j)^{SL(W_j)} = \mathbf{C}\omega_j^2 \oplus \det W_j \oplus \det W_j^\vee,$$

and the invariant spaces in degrees one and three vanish. Applying $\bigwedge^4(U_1 \oplus U_2)$ gives exactly the three divisor products $\omega_1^2, \omega_1\omega_2, \omega_2^2$ and the four determinant lines $\det W_1, \det W_1^\vee, \det W_2, \det W_2^\vee$. Here ω_1, ω_2 span $\mathrm{NS}(B_8)_{\mathbf{C}}$; the four determinant lines are $W_E(B_8)_{\mathbf{C}}$. They lie in distinct summands of the exterior-power decomposition, so

$$\mathrm{Hdg}^2(B_8)_{\mathbf{Q}} = D^2(B_8) \oplus W_E(B_8), \qquad \dim_{\mathbf{Q}} D^2(B_8) = 3, \quad \dim_{\mathbf{Q}} W_E(B_8) = 4.$$

Thus the codimension-two Hodge conjecture for B_8 is equivalent to algebraicity of $W_E(B_8)$: all classes in $D^2(B_8)$ are products of divisor classes, and the displayed decomposition leaves no other codimension-two Hodge classes. By the structural theorem and (7.4), this is also equivalent to the classical Hodge conjecture holding for every self-power B_8^n.

8. Abdulali's generalized Hodge conjecture corollary

The argument of this section is a direct application of published results and claims no new mechanism. The deduction of the generalized Hodge conjecture from the classical Hodge conjecture for all self-powers is the Hazama–Abdulali paradigm; see Hazama [11] for the stably nondegenerate case (whose Type I/II hypothesis does not apply to B_8), Abdulali [18, Theorem 4.1] for the corrected semisimple, no-Type-III case used here (with the corrections to [10, Theorem 5.1] recorded in [18, Appendix B.1]), and Abdulali [12] for a short proof in the CM setting.

Corollary

Assume (2.1). Then for every $n \geq 1$, the fixed self-power B_8^n satisfies the generalized Hodge conjecture.

Proof Fix n and put $X = B_8^n$. Since $\mathrm{End}^0(B_8) = E$, one has

$$H^1(X, \mathbf{Q}) = V \otimes \mathbf{Q}^n, \qquad \mathrm{End}^0(X) = M_n(E). \tag{8.1}$$

The matrix units in $M_n(E)$ are Hodge endomorphisms. Hence $Hg(X)$ centralizes them, while the centralizer of $M_n(E)$ on $V \otimes \mathbf{Q}^n$ is the diagonal copy of $GL_E(V)$. Restriction to any summand and functoriality of the Hodge group therefore give $Hg(X) = \Delta Hg(B_8)$. For the product polarization on X, the Rosati involution on $M_n(E)$ is conjugate transpose, and its centralizer in the symplectic group is the diagonal unitary group. Thus

$$L(X) = \Delta \,\mathrm{Res}_{E^+/\mathbf{Q}}\, U_E(V, h), \qquad Hg(X) = \Delta Hg(B_8), \tag{8.2}$$

so $L(X)^{\mathrm{der}} = Hg(X)$ by (1.3). In particular X is PEL-type and its Hodge group is semisimple. The only simple isogeny type is the Type-IV factor B_8, so X has no Type-III simple factor. Thus X meets the hypotheses of Abdulali's corrected Theorem 4.1 [18]: it is PEL-type, has semisimple Hodge group, and has no Type-III simple factors. Finally $X^r = B_8^{nr}$, and Section 7 proves the classical Hodge conjecture for every $r \geq 1$, conditional on (2.1). The corrected theorem therefore gives the generalized Hodge conjecture for X (indeed for every power of X) once (2.1) is established from H1–H4.

9. Scope and limitations

The conditional transfer in this paper concerns one fixed Fermat character and one fixed Prym model. Its source block is obtained by applying the graph correspondence and rational orbit projector to the algebraic cubic-scroll lattice (with Aoki's construction as an independent source proof), then using the displayed adjoint and small-diagonal correspondences. Section 6.2 gives a finite scroll-indexed spanning family. Proposition 6.1 proves full character support for

each source scroll and hence gives a projected witness for (6.12). Proposition 6.2 proves that the order-five- averaged scroll family spans the target, so four independent averaged cycles exist; it does not specify their indices. An individual unaveraged scroll image may still vanish after projection. Four one-factor endomorphism correspondences give a basis from any nonzero member. The results on B_8^n are stated under H1–H4, in the single dependency chain described in the introduction. An alternative Abel–Fourier route is not used. The theorem does not address the general Fermat generalized Hodge conjecture, the general Hodge conjecture, or algebraicity of all Hodge classes on X_{33}^4.

Relation to the literature on Fermat varieties

For Fermat hypersurfaces themselves, Kang has stated the generalized Hodge conjecture for X_m^n of dimension three or four and arbitrary degree m [26, Corollary 3.2], by a refined motivic dimension argument resting on [27, Proposition 2.2]. We do not rely on that statement, and we assert no priority relative to it: our source algebraicity comes from the level-66 lift of Jumagulov [28, Theorem 1.4], which rests on Aoki's theorem [20], and the assertions proved here concern the Prym factor B_8 and its powers rather than X_{33}^4.

Jumagulov [28] presents a computer-assisted proof of the Hodge conjecture for every Fermat fourfold of odd degree $m \leq 199$ (Theorems 1.1 and 1.5). His Theorem 1.4 (July 2026), which predates the present paper, gave the level-66 lift of the character $w = (1, 4, 16, 22, 25, 31)$, whose orbit contains $\alpha = 7w$, together with its Lefschetz, da Silva and Aoki closure; the source argument of Section 3 follows that lift, and we claim no priority for it. We do not use his census as an input. His Theorem 1.6 and Corollary 1.7 constrain the field of definition of any cycle projecting onto this orbit; see Section 3. The question addressed here is the conditional transfer to the fixed B_8, not Fermat-side algebraicity.

Rinaldi [29, Theorem A and §4] gives a further preprint claim for every Fermat fourfold of degree at most 100 except 70. At degree 33, the paper supplies a printed level-66 character identity and descent of the exceptional orbit. We checked that displayed identity, but not the full census or its checker. This is additional source-side evidence for the same Fermat character orbit; it gives no correspondence to the fixed B_8 and is not used as an input here.

Miranda, Movasati, Rufino, and Villaflor [30, Theorem 1.1] claim the Hodge conjecture for all Fermat varieties of degree below 65 except degrees 44, 51, and 52, which includes X_{33}^4. Their reduction explicitly uses Kang's Fermat-fourfold theorem. It therefore does not independently close the Kang input under review here, and it gives no transfer to the fixed B_8.

Bhattacharjee's v1.7.0 preprint (11 October 2026) [31, Theorem B] claims the Hodge conjecture for every odd-degree Fermat fourfold, including X_{33}^4. For $m = 33$, its exceptional orbit O_{33} is resolved by an identity that the paper explicitly attributes to Jumagulov [28, Theorem 1.4, p. 23]. It therefore does not give an independent verification of this degree-33 source character, and Theorem B supplies no transfer to the fixed ramified Prym B_8. We continue to use the displayed Aoki–da Silva source argument and treat the target transfer as conditional on H1–H4.

Favero–Kelly [23, §4.3, Theorem 4.8] independently claim the Hodge conjecture for X_{33}^4, using a pullback from a cubic fourfold; their character $(19, 7, 13, 10, 28, 22)$ is a permutation of $\alpha = (7, 10, 13, 19, 22, 28)$. In the posted v1, however, equation (4.1) has final term x_3^3, so the cubic is a cone with vertex $[0 : 0 : 0 : 0 : 0 : 1]$. The monomial exponents in the stated map (4.4) instead give $y_0^2 y_1 = x_0^{33}, \ldots, y_4^2 y_0 = x_4^{33}$, and $y_5^3 = x_5^{33}$; thus the stated map lands in the cubic only if its final term is y_5^3, not y_3^3. The arXiv record still lists only v1 as of 16 October 2026. This is a concrete gap in the proof as posted, not a disproof of the theorem. Section 6.2 is an elaboration of their strategy: it uses the same monomial map, with the corrected smooth cubic (final term y_5^3), and the Billi–Grossi–Marquand scroll lattice to obtain the full source block.

Relation to concurrent rStation work

Three published rStation papers and one forum reduction are close to the present work. The Legendre-symbol detector and 2-primary Weil-field obstruction of [37] determine the minimal Weil fields of the exceptional Fermat classes $\xi_{44}, \xi_{51}, \xi_{52}$ and prove new Fermat cases by exhaustive computation; they work on the Fermat (source) side and give no transfer to an abelian variety. The Lefschetz barrier of [38] shows that classes generated by imaginary-quadratic Weil classes cannot reach certain determinant classes; the quartic-CM determinant $W_E(B_8)$ is of exactly this kind, which is why a transfer from a Fermat source is used here. The norm obstruction of [39] concerns the degree-44 class in the K3 representation. Most closely, the forum note [40] reduces the degree-44 class to a determinant on an explicit ten-dimensional Prym factor over the quartic field $\mathbf{Q}(i, \sqrt{-11})$, and identifies Markman's semiregularity step [1] as the

missing input. The present paper treats the analogous degree-33 situation over a different quartic field and replaces the secant-sheaf route by an explicit small-diagonal correspondence (Section 6), conditional only on the fixed model quotient inputs H1–H4.

Relation to the recent dimension-four results

The transfer mechanism used here — algebraicity on a Fermat source implies algebraicity of a determinant substructure on an associated generalized Prym variety — is not original to this paper. It is the mechanism of Schoen's reduction for cyclic covers of $\mathbf{P}^v$ branched along $v+2$ hyperplanes [8]; the branch data fixed above do not satisfy the quasi-simple hypothesis under which [8, Theorem 3.4] concludes directly, which is why the explicit chain of Sections 3–6 is carried out, but no priority is claimed for the implication itself.

It may be worth recording precisely why neither route available in [8] applies to the fixed data of Section 1, since both routes are genuinely more general than the worked examples of [8, §6]. In the notation of [8, §1, §3] our branch data are $q=0$, $s=6$, $m=33$, with m-tuple $\alpha=(7,10,13,19,22,28)$, so that $h=2q-2+s=4$. No two entries of α sum to 0 modulo 33, so the largest admissible r in Schoen's setup is $r=1$, giving $v=s-2r=4$. But [8, §3] additionally assumes $r>1$ when $q=0$; hence its correspondence is not stated for this branch tuple. On the other hand the multiplicity $\nu_\chi = h/2 - s/2 + m^{-1}\sum_i \langle -n\alpha_i\rangle_m$ equals 3 for every primitive character $\chi=n\varepsilon$, whence by [8, (1.2)] the Hodge width is

$$w = \max_{n\in(\mathbf{Z}/33)^*} \left| s - 2m^{-1}\textstyle\sum_i \langle -n\alpha_i\rangle_m \right| = 0.$$

Thus $w=0<4=v$: the quasi-simple hypothesis $v=w$ of [8, Theorem 3.4] also fails. Remark 3.5 describes a sufficient reduction from the Fermat source, but it follows the same §3 setup and is not a stated result for the excluded $r=1, q=0$ case. Its text invokes the full Fermat Hodge conjecture as a sufficient input; here we establish only the selected primitive character block by the level-66 identity. Applying that remark to this single block would require an additional character-level extension of Schoen's correspondence. Sections 3–6 instead give a fixed model correspondence and prove the required map directly; no novelty is claimed for the general Fermat-to-Prym transfer mechanism.

The Weil-class problem in dimension four is now settled. Markman proved algebraicity of the Hodge–Weil classes on abelian fourfolds of Weil type [1,2], which together with the Moonen–Zarhin structure theorem [3,4] yields the Hodge conjecture for all abelian varieties of dimension at most four; Floccari–Fu [5] and Varesco [6] obtain the Hodge conjecture for all powers of discriminant-one Weil fourfolds. The division of labour between [2] and [1] is worth stating precisely, since it also records how much of the four-dimensional problem was already due to Schoen: [2, Theorem 1.5] establishes algebraicity for abelian fourfolds of Weil type with multiplication by an arbitrary imaginary quadratic field but of discriminant one, the arbitrary-discriminant statement being supplied by the later secant-sheaf work [1]; and, as Markman records in the same place, the case $K=\mathbf{Q}(\sqrt{-3})$ had been obtained for arbitrary discriminant by Schoen [32], and the case $K=\mathbf{Q}(i)$ of discriminant one in [15]. All of these results are four-dimensional and concern determinant subspaces attached to an imaginary quadratic field. The object treated here has dimension eight, $\mathrm{End}^0(B_8)=E$ a quartic CM field, and $\dim_E H^1(B_8,\mathbf{Q})=4$; none of the cited results applies to it. Conversely, we make no claim about any eight-dimensional object other than this fixed B_8. See also van Geemen [33] for the surrounding Weil-class context in dimension four.

Nor is the result a direct instance of Aoki's Abelian-side theorem [34, Theorems 2.4–2.5]. That theorem concerns simple full-CM factors of Fermat Jacobians and their powers. Here

$$\mathrm{End}^0(B_8)=E, \qquad [E:\mathbf{Q}]=4, \qquad \dim_E H^1(B_8,\mathbf{Q})=4,$$

so B_8 is not of full CM type; in particular it cannot be one of those simple factors. It is not isogenous to a nontrivial power of one either, since that would give a matrix endomorphism algebra, whereas $\mathrm{End}^0(B_8)=E$ is a field. Aoki's theorem supplies relevant source algebraicity but does not itself produce the quartic-E determinant class on B_8.

The 6 October 2026 OpenAI preprint [7, "Faithful cyclic determinants"] states that for a connected cyclic cover $C\to B$ with faithful character χ, the determinant line $D(H^1(C)_\chi)$ is supplied on its curve slots by rational algebraic correspondences from powers of $\mathrm{Jac}(F_m^1)$, where m is the deck-group order. For the degree-33 map $C\to\mathbf{P}^1_z$ defined in Section 1, $(g(B),r,m)=(0,6,33)$, so $\dim H^1(C)_\chi = 2g+r-2=4$, and the lemma applies to every primitive character. This fixed-cover supply statement is separate from that preprint's tensor-Hodge-generic theorem. Its use to deduce algebraicity of Hodge classes depends on the preprint's separate CM-Hodge input; it gives no correspondence to B_8. We do not

use [7]: Section 3 records the full source block independently, via the level-66 lift of [28, Theorem 1.4], while Section 6.2 proves the same source block by pulling back the primitive cubic $(2,2)$-space onto the full Fermat unit orbit. The transfer needs only that source line and the specific curve-to-Prym small-diagonal correspondence of Sections 5–6.

The same preprint's full-variation theorem cannot be applied to this particular marked point. Under our graph-correspondence convention, the $H^1(C) \otimes H^1(C)$ Künneth component of $\Gamma_{\rho^{-1}}$, identified with an endomorphism tensor by the polarization, represents $R = \rho^*$. It is a rational tensor of type $(1,1)$ at the displayed curve. Its Hodge locus has finite image in the moduli of the six labelled branch points. Indeed, if it were Hodge at a fiber, R would be an integral symplectic Hodge automorphism of $J(C_s)$. Riemann–Hurwitz gives $g(C_s) = 59$; strong Torelli (Milne, *Abelian Varieties*, Theorem 12.1) would then give $R = \pm f^*$ for an automorphism f of C_s. Writing $D = \delta^*$, the relation in Section 1 gives $RDR^{-1} = D^{25}$, since $4^{-1} = 25 \pmod{33}$. The sign cancels in conjugation, so $(f^{-1}\delta f)^* = D^{25} = (\delta^{25})^*$. Faithfulness of the curve-automorphism action on its Jacobian yields $f^{-1}\delta f = \delta^{25}$, or $f\delta f^{-1} = \delta^4$. Thus f normalizes the deck group and descends to a nonidentity automorphism g of $\mathbf{P}^1$; if $g = 1$, then f is a deck transformation and commutes with δ, a contradiction. If a branch value has monodromy exponent a, the descended map carries it to exponent $4a \pmod{33}$. The six exponents $7, 28, 13, 19, 10, 22$ are distinct; multiplication by 4 cycles the first five and fixes 22. Hence g^5 fixes all six branch values, so $g^5 = 1$, and g has order five. After conjugating, write $g(z) = \zeta_5^q z$, with $q \in (\mathbf{Z}/5)^\times$, put the fixed branch value at ∞, and put the branch value of exponent 7 at 1. The remaining labelled branch values are then forced to be

$$P_{28} = \zeta_5^q, \quad P_{13} = \zeta_5^{2q}, \quad P_{19} = \zeta_5^{3q}, \quad P_{10} = \zeta_5^{4q}.$$

Thus at most four labelled configurations occur in the coarse branch moduli. The Hodge locus is zero-dimensional on the full marked variation (with at most countably many marking lifts). The displayed symmetric curve is one such point, but the tensor is not Hodge throughout the variation; this excludes it from the preprint's tensor-Hodge-generic locus.

The result is also distinct in scope from recent Prym and secant-sheaf constructions. Van Geemen–Verra [35] realize a six-dimensional family of quaternionic abelian eightfolds as Prym varieties of unramified quaternionic covers and use Schoen's method to prove algebraicity of their quaternionic Hodge classes. Their definite quaternion endomorphism algebra and unramified covering construction do not contain the commutative quartic-CM, ramified fixed factor treated here. Patel–Zhang's Theorem 1.1 [36] assumes an étale finite abelian cover $C \to C'$ with $g(C') \geq 2$; it does not treat the ramified Jacobian-with-modulus situation used here. Markman's Theorem 1.1.2 [1] is conditional on algebraicity being preserved along the relevant flat deformation. The higher-degree semiregularity problem is left open there, and in the quartic CM case with $[F : \mathbf{Q}] = 2$ the required algebraicity is not asserted. These hypotheses do not replace the finite-quotient and small-diagonal tensor–cup construction proved in this paper. Floccari–Fu [5] prove the Hodge conjecture for discriminant-one Weil fourfolds and all their powers by an OG6 construction. Their four-dimensional, imaginary-quadratic setting and geometric correspondence are different from the ramified quartic-CM Prym factor and finite-quotient transfer considered here.

The Aoki factorization route still leaves $Q = (1, 25, 44, 62)$ as its exceptional level-66 surface character, while β, γ occur in the standard Aoki–Shioda stratum [20,13]; this limits explicitization by that route, not the existence of the source block proved above. The cited Billi–Grossi–Marquand result shows that the 6270 cubic-scroll classes generate the cubic algebraic lattice; Section 6.2 transports their primitive projections to a finite spanning family on B_8. Proposition 6.1 proves full character support for each source scroll, but the family-level span—not individual nonvanishing—supplies a nonzero target seed under H1–H4. Four one-factor endomorphism correspondences give a basis from any nonzero member. Coordinate equations for the scrolls and transferred cycles remain unavailable. The Tate-normalized Jacobi character $u_Q(67) = \zeta_{66}^{58} \neq 1$ explains why an unweighted Galois trace does not provide the missing explicit divisor in the Aoki route.

Reproducibility

The discrete facts used in the proof are exact calculations: Frobenius polynomials and the number-field elimination for $\mathrm{End}^0(B_8)$, the level-66 identity, the coordinates of the common quotient and its field of definition, the character computations for the projectors, and the normalization 5^{-3}. The programs are in the supplement, under `calculations/bundles/`. They need Python $\geq$ 3.10 with `sympy`, SageMath (for the Frobenius polynomials) and PARI/GP (for the number-field elimination). Their role is only to confirm these finite statements; the geometric arguments are the ones written above.

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